Emotional Dysregulation as the Core of Transdiagnostic Psychopathology

Chris Davis[1,*], Claude Opus 4.6[2], ToolUniverse[3]

[1]Independent Researcher, Medford, MA, USA [2]Anthropic, San Francisco, CA, USA (model: claude-opus-4-6, 1M context) [3]Harvard MIMS Lab, Boston, MA, USA (ToolUniverse v1.1.5)

[*]Correspondence: chris@bigbio.ai

ABSTRACT

Emotional dysregulation—the inability to modulate affective states in a context-appropriate manner—operates as a transdiagnostic mechanism across borderline personality disorder, post-traumatic stress disorder, depression, attention-deficit/hyperactivity disorder, and substance use disorders. This review synthesizes functional neuroimaging, genome-wide association studies, molecular neurobiology, and randomized clinical trials to propose an integrated neurocircuit model. Convergent evidence implicates deficient prefrontal–amygdala connectivity, rather than amygdala hyperactivation, as the core neural phenotype. Genome-wide analyses of neuroticism in over 500,000 individuals reveal a polygenic architecture that overlaps substantially with depression and anxiety. Functional enrichment of eight candidate genes converges on behaviour regulation, neurotransmitter catabolism, and hypothalamic–pituitary–adrenal axis function. Epigenetic studies demonstrate that adversity inscribes partially reversible methylation signatures onto these same genes, linking environment to circuit vulnerability. Dialectical behaviour therapy emerges as the most robustly validated intervention, with recent randomized trials extending its efficacy to autism, bipolar disorder, and chronic pain. Emerging circuit-targeted approaches—repetitive transcranial magnetic stimulation combined with cognitive restructuring, intranasal oxytocin—offer mechanistically principled alternatives. Emotional dysregulation constitutes a unifying framework that should reorient classification and treatment toward dimensional, circuit-based targets.

INTRODUCTION

Clinicians diagnose emotional disorders in categories—major depressive disorder, borderline personality disorder, post-traumatic stress disorder, attention-deficit/hyperactivity disorder—yet the patients who carry these labels share a single, recurring failure: they cannot modulate the intensity, duration, or expression of their emotional responses to match the demands of the moment. This mismatch between categorical nosology and dimensional reality has plagued psychiatry for decades. Comorbidity rates exceed 50% across mood, anxiety, and personality disorder diagnoses, and factor-analytic studies consistently extract a general "internalising" dimension that straddles every boundary the *Diagnostic and Statistical Manual* draws.[1] If the categories carve nature at its joints, the joints should not bleed this freely. The bleeding suggests that the categories are wrong—not in their clinical utility, which remains indispensable for treatment guidelines and insurance codes, but in their claim to reflect discrete biological entities. What biology offers instead is a continuum of regulatory capacity that varies across individuals, contexts, and developmental epochs, and that breaks down in characteristically different ways depending on which circuits and genes are compromised.

The construct that cuts across every diagnostic joint is *emotional dysregulation*: a transdiagnostic deficit in the capacity to monitor, evaluate, and modify emotional reactions in the service of adaptive goals. Before Gross's foundational work, "emotion regulation" was a loosely deployed umbrella term that conflated coping, defence mechanisms, mood repair, and self-control without specifying which psychological operations these constructs shared or how they differed. Gross resolved this ambiguity by proposing the process model of emotion regulation, which decomposed the regulatory act into five sequential stages: situation selection, situation modification, attentional deployment, cognitive change, and response modulation.[2] Each stage corresponds to a distinct point in the emotion-generative trajectory at which a person can intervene. Situation selection and situation modification operate on the external world before an emotional episode begins—choosing whether to attend a funeral or rewriting the seating chart to avoid a hostile relative. Attentional deployment and cognitive change operate on internal representations once the episode is underway—directing gaze away from distressing stimuli or reinterpreting a critic's feedback as an opportunity for growth. Response modulation operates on the physiological and behavioural output after the

emotion has been fully generated—deep breathing to slow heart rate, or suppressing a facial expression of anger. This temporal ordering matters because antecedent-focused strategies, which act before the full emotional response has been assembled, prove less metabolically expensive and more effective than response-focused strategies, which attempt to suppress an emotion already in flight.[2] The process model thus transformed emotion regulation from a folk-psychological concept into a mechanistically specified sequence with measurable decision points, each amenable to experimental manipulation and each mapping onto identifiable neural substrates.

John and Gross sharpened this five-stage architecture into two prototypical strategies that have since anchored thousands of empirical studies.[3] Cognitive reappraisal—reinterpreting a situation to alter its emotional meaning—reduces negative affect, preserves social connection, and costs little in working memory. Expressive suppression—inhibiting the outward display of an emotion without changing the internal experience—fails to reduce subjective distress, impairs episodic memory for the triggering event, and erodes interpersonal rapport over repeated interactions. The asymmetry between these two strategies is not merely quantitative; it is qualitative. Reappraisal alters the emotional trajectory early, before autonomic and endocrine cascades reach full amplitude, while suppression leaves the full physiological response intact and adds a second metabolic burden—the effortful inhibition of expressive behaviour—on top of it. Habitual reappraisers show lower sympathetic nervous system activation during social stress, lower cortisol reactivity during laboratory challenges, and faster cardiovascular recovery after negative emotional episodes. Habitual suppressors show the reverse on every measure.[3] Individual differences in habitual strategy use therefore predict well-being more reliably than the frequency or intensity of negative life events: people who habitually reappraise report higher life satisfaction, fewer depressive symptoms, and stronger social bonds, while people who habitually suppress report lower satisfaction, more symptoms, and progressive social withdrawal. The reappraisal–suppression axis offers a dimensional metric of regulatory health that transcends diagnostic categories. A patient diagnosed with PTSD who habitually reappraises may function better than a patient with no formal diagnosis who habitually suppresses—a paradox that categorical nosology cannot accommodate but that dimensional models predict and explain.

Building on this dimensional logic, Linehan's biosocial theory located the developmental origins of severe dysregulation at the intersection of biological vulnerability and environmental invalidation.[4] Children born with heightened emotional sensitivity, rapid emotional escalation, and slow return to baseline—a temperamental triad mediated by amygdala reactivity and prefrontal maturation rates—develop adaptive regulation only when their caregiving environment acknowledges, labels, and coaches emotional experience. When the environment instead dismisses, punishes, or ignores emotional expression, the child never acquires the cognitive reappraisal strategies that Gross's model identifies as adaptive. The result is a regulatory repertoire dominated by suppression, avoidance, and impulsive action—strategies that provide momentary relief at the cost of long-term escalation. Crowell and colleagues formalised this account as a developmental cascade: heritable neurocognitive vulnerabilities interact with invalidating environments across sensitive periods to produce the affective instability, identity disturbance, and interpersonal chaos characteristic of borderline personality disorder.[4] Crucially, however, the same cascade—with different weightings of vulnerability and invalidation—generates the dysregulation observed in PTSD, recurrent depression, ADHD, and substance-use disorders. In PTSD, the vulnerability is a sensitised threat circuit and the invalidation is the trauma itself; in ADHD, the vulnerability is a sluggish reward circuit and the invalidation is an environment that demands sustained effortful control; in substance-use disorders, the vulnerability is an over-responsive dopaminergic system and the invalidation is early exposure to substances during a period of prefrontal immaturity. The biosocial model is therefore not a theory of one diagnosis; it is a theory of one mechanism expressed across many diagnoses.

This transdiagnostic convergence motivated the National Institute of Mental Health to abandon categorical funding priorities in favour of the Research Domain Criteria (RDoC) framework.[5] RDoC organises psychopathology along five dimensional constructs—negative valence systems, positive valence systems, cognitive systems, social processes, and arousal/regulatory systems—each analysed across seven units of analysis that span genes, molecules, cells, circuits, physiology, behaviour, and self-report. Emotional dysregulation sits at the intersection of the negative valence and arousal/regulatory domains, precisely where circuit-level and genetic evidence converge most densely. Within the negative valence domain, dysregulation maps onto the "sustained threat" and "frustrative nonreward" constructs; within the arousal/regulatory domain, it maps onto the "arousal" and "circadian rhythms" constructs that govern the temporal dynamics of emotional responding. By replacing the question "Which disorder does this patient have?" with "Which regulatory dimensions are disrupted, at which units of analysis, and to what degree?", RDoC reframes emotional dysregulation as a measurable, mechanistically decomposable target rather than a loose clinical descriptor. The framework also encourages researchers to study the full range of regulatory variation—from resilient individuals who regulate effortlessly through subclinical populations who struggle under stress to clinical populations who cannot regulate at all—rather than comparing categorical "cases" with categorical "controls" and discarding the dimensional information that lies between them.

Phenotype ontologies offer a complementary strategy for escaping categorical constraints. The Human Phenotype Ontology (HPO) term `HP:0100851`—Abnormal emotional state—subsumes twelve descendant phenotypes including anhedonia (`HP:0012154`), irritability (`HP:0000737`), dysphoria (`HP:0100020`), emotional lability (`HP:0000712`), anger (`HP:0031473`), and apathy (`HP:0000741`).[6] Each descendant maps to distinct neural circuit signatures, genetic architectures, and pharmacological response profiles, yet all share the parent node. This hierarchical structure mirrors the empirical reality that dysregulation is not one phenotype but a family of phenotypes organised by severity, valence, and temporal dynamics. Anhedonia involves a failure to generate positive affect; irritability involves a failure to inhibit negative affect; emotional lability involves rapid oscillation between affective states; apathy involves a failure to generate any affect at all. These facets can co-occur, dissociate, and respond differently to treatment within the same individual. Mapping clinical observations onto

HPO terms enables cross-study harmonisation, large-scale phenome-wide association studies, and the precise specification of which *facet* of dysregulation a given gene, circuit, or intervention targets.

The present review exploits this dimensional, multi-level architecture. We begin by mapping the neural circuitry that implements—and fails to implement—emotion regulation, drawing on converging evidence from structural and functional neuroimaging, lesion studies, and causal perturbation experiments. We then trace the genetic architecture that tunes this circuitry, moving from genome-wide association signals through gene-expression landscapes to rare-variant convergence on shared synaptic pathways. Next, we examine developmental trajectories in which genetic risk and circuit maturation interact with environmental exposures to crystallise stable patterns of dysregulation. Finally, we evaluate therapeutic interventions—pharmacological, psychotherapeutic, and neuromodulatory—through the lens of the circuits and genes they engage. Throughout, we anchor our analysis in the RDoC framework and HPO phenotype hierarchy, treating emotional dysregulation not as a symptom of a category but as a measurable dimension that demands mechanistic explanation at every level of biological organisation.

NEUROCIRCUIT MODEL OF EMOTIONAL DYSREGULATION

The prefrontal–amygdala axis has dominated neurocircuit models of emotion regulation for two decades, yet the field's understanding of *how* this axis fails has undergone a fundamental revision. Early formulations cast the amygdala as a threat-detection alarm and the prefrontal cortex as its top-down brake: dysregulation, on this account, arose from an overactive amygdala that overwhelmed prefrontal inhibition. This "see-saw" model was intuitive, clinically appealing, and wrong in its central claim. Silverman and colleagues overturned the simple story in a large-sample ($N > 500$) study that parsed neuroticism—the personality trait most strongly and consistently associated with emotional dysregulation—into its constituent circuit signatures.[7] They found that neuroticism predicted neither amygdala hyperactivation nor prefrontal hypoactivation in isolation; instead, it tracked *deficient functional connectivity* between the amygdala and ventromedial prefrontal cortex (vmPFC). The amygdala generated threat signals at normal amplitude, but those signals failed to propagate to the prefrontal regions that would contextualise, reappraise, and ultimately down-regulate them. Dysregulation, this finding revealed, is a disconnection disorder, not an excitation disorder—a failure of communication between nodes, not a failure of any single node.

Kredlow and colleagues extended this connectivity-centric view to PTSD by decomposing prefrontal–amygdala interactions into three temporally distinct phases: fear acquisition, fear extinction, and fear regulation.[8] During acquisition, patients with PTSD showed intact amygdala signalling—they learned the threat contingency as rapidly as trauma-exposed controls, and their amygdala BOLD responses to conditioned stimuli were indistinguishable from those of healthy participants. During extinction, however, vmPFC failed to suppress the acquired fear response: the safety signal was encoded but could not override the threat representation. During active regulation—when participants were instructed to reappraise the threatening stimulus—dlPFC failed to recruit vmPFC to implement the cognitive transformation that reappraisal demands. This three-phase dissection clarified a clinical puzzle that had resisted explanation for years: why patients with PTSD *can* learn new safety signals in the controlled laboratory environment yet *cannot* deploy them in ecologically valid contexts where competing demands strain prefrontal resources. The acquisition circuit works; the extinction and regulation circuits do not. The therapeutic implication follows directly—interventions must target connectivity during extinction and regulation, not amygdala reactivity during acquisition, and exposure-based therapies must be augmented with strategies that strengthen the prefrontal–amygdala pathway rather than simply re-presenting the feared stimulus.

Depression presents a complementary pattern that further fragments the monolithic "PFC dysfunction" narrative. Pizzagalli and Roberts reviewed convergent evidence from task-based fMRI, resting-state connectivity analyses, and computational reinforcement-learning models to show that prefrontal disruption in depression is subtype-specific.[9] Anhedonic depression—characterised by diminished pleasure, reduced motivation, and blunted reward learning—maps to reduced connectivity between medial PFC and ventral striatum, reflecting an impaired reward-prediction error signal that starves the cortex of the positive-valence information needed to guide goal-directed behaviour. Anxious depression—characterised by hypervigilance, somatic tension, and ruminative worry—maps to exaggerated connectivity between the dorsal anterior cingulate cortex (dACC) and the amygdala, reflecting an overtuned threat-monitoring loop that biases attention toward negative stimuli at the expense of neutral and positive alternatives. Melancholic depression—characterised by psychomotor retardation, diurnal mood variation, and vegetative signs—maps to reduced dlPFC activation during cognitive control tasks, reflecting a depleted regulatory resource that can neither reappraise threat nor amplify reward. No single prefrontal abnormality captures "depression"; the diagnostic label masks at least three circuit-level disorders, each with distinct pharmacological and psychotherapeutic response profiles. This heterogeneity explains decades of inconsistent neuroimaging findings and argues forcefully for the dimensional, circuit-specified approach that RDoC demands. It also suggests that treatment selection should be guided not by the syndromal diagnosis but by the circuit subtype—a strategy that emerging imaging-guided treatment algorithms are beginning to test.

Beyond mood and anxiety disorders, the prefrontal–subcortical axis also governs the dysregulated reward pursuit that characterises addictive behaviours. Brand and colleagues proposed the Interaction of Person–Affect–Cognition–Execution (I-PACE) model, which has since attracted over 1,600 citations, to formalise how fronto-striatal imbalance drives compulsive engagement with substances and behavioural rewards alike.[10] In their framework, affective and cognitive responses to triggers interact within a prefrontal–striatal loop:

ventral striatal signals encode the incentive salience of a cue, while dlPFC and inferior frontal gyrus (IFG) compute the long-term costs of acting on that cue. When prefrontal inputs weaken—through chronic stress, substance-induced neurotoxicity, or trait-level impulsivity that reflects developmental prefrontal immaturity—the striatal signal wins, and the person pursues the reward despite foreseeing its negative consequences. The I-PACE model thus extends the disconnection framework from threat regulation (amygdala–PFC) to reward regulation (striatum–PFC), revealing a shared architectural principle: dysregulation arises whenever a subcortical signal is generated at normal or elevated intensity but the prefrontal circuit that should contextualise and constrain that signal is structurally or functionally disconnected. This principle unifies threat-related and reward-related forms of dysregulation under a single computational logic—top-down contextualisation fails, and the subcortical signal drives behaviour unopposed.

Recent causal evidence has elevated this correlational architecture to mechanistic status. Niu and colleagues used concurrent transcranial magnetic stimulation and functional MRI (TMS–fMRI) to demonstrate that single-pulse stimulation of the left dlPFC produced immediate, measurable changes in amygdala blood-oxygen-level-dependent (BOLD) signal in healthy participants.[11] The same stimulation protocol, applied to individuals with subthreshold depression, *failed* to modulate amygdala activity: the dlPFC was activated, but the downstream signal did not arrive. This causal probe confirmed what correlational connectivity analyses had inferred: the dlPFC-to-amygdala regulatory pathway is not merely associated with mood state but is causally necessary for effective emotional down-regulation, and this causal link degrades before clinical thresholds are crossed. Subthreshold depression, on this evidence, is already a circuit disorder, not merely a risk factor for one. The clinical implication is that intervention at the subthreshold stage—when the pathway is weakened but not yet severed—may prevent the transition to full syndromal disorder, a possibility that current neurostimulation trials are actively exploring.

Meiering and colleagues approached the same circuitry from the trait-vulnerability direction, using high-resolution fMRI during an emotional face-processing task to show that neuroticism predicts increased *functional* connectivity among the amygdala, hippocampus, and medial PFC during the processing of negative emotional stimuli.[12] At first glance, this finding appears to contradict the Silverman result of *deficient* amygdala–vmPFC connectivity in high-neuroticism individuals. The resolution lies in the critical distinction between regulatory connectivity and reactive connectivity. Silverman measured connectivity during explicit emotion regulation—participants were instructed to down-regulate their emotional response; Meiering measured connectivity during passive emotional processing—participants simply viewed the stimuli without regulatory instruction. Neuroticism, therefore, involves a double dissociation: heightened amygdala–hippocampal–PFC coupling during the initial encoding of emotional events, which amplifies the subjective salience of negative experience and strengthens the contextual memory trace that binds the negative emotion to its situational triggers, and weakened amygdala–vmPFC coupling during the subsequent attempt to regulate that experience, which prevents effective down-regulation. The neurotic brain, in short, amplifies the emotional signal through enhanced reactive connectivity and then disconnects the regulatory brake through impaired top-down control. This double dissociation explains why highly neurotic individuals report both more intense negative emotions *and* more difficulty managing those emotions—features that a simple hyperactivation or hypoactivation model cannot capture.

These macro-circuit findings acquire finer anatomical grain when mapped onto the Allen Human Brain Atlas, which parcellates the human amygdala into 88 sub-regions spanning the lateral nucleus, basal nucleus, accessory basal nucleus, central nucleus, cortical nuclei, and their laminar subdivisions.[13] The lateral nucleus receives the densest cortical input—from auditory, visual, and polymodal association cortices—and serves as the primary gateway for sensory information entering the amygdala. The central nucleus projects to brainstem effector regions, including the periaqueductal grey, parabrachial nucleus, and dorsal vagal complex, that generate the autonomic, endocrine, and motor components of the emotional response. The basal nucleus bridges these input and output streams while maintaining dense reciprocal connections with vmPFC, orbitofrontal cortex, and the nucleus accumbens, positioning it as the critical hub through which prefrontal cortex exerts its regulatory influence. The connectivity deficits reported by Silverman, Kredlow, and Niu likely localise to the basal nucleus–vmPFC pathway; the reactive connectivity amplification reported by Meiering likely engages the lateral nucleus–hippocampal pathway. Future studies that combine ultra-high-field imaging (7T and above) with atlas-informed parcellation will be positioned to test this sub-regional hypothesis directly, moving the field from treating the "amygdala" as a monolithic computational node to recognising it as a differentiated structure whose sub-circuits serve distinct regulatory, reactive, and output-generation functions.

How, then, do these circuit disruptions relate to the adaptive regulation that Gross's process model describes? Troy and colleagues addressed this question by proposing an affect-regulation framework for psychological resilience.[14] In their model, positive reappraisal—the cognitive strategy of reinterpreting a negative event as an opportunity for growth, learning, or meaning—serves as the central mechanism linking prefrontal circuit integrity to adaptive outcomes. Resilient individuals, Troy and colleagues showed, do not experience less adversity and do not generate weaker amygdala responses to that adversity; rather, they recruit vmPFC and dlPFC more effectively during reappraisal, which transforms the valence of the amygdala signal before it propagates to downstream effector circuits. The reappraised signal still carries information about the threatening stimulus, but that information is now contextualised by a narrative of agency, growth, or meaning that reduces its autonomic impact and frees attentional resources for goal-directed action. Resilience, on this account, is not a fixed trait that resides in a brain region; it is a dynamic process that emerges from the efficiency of a circuit—precisely the amygdala–vmPFC–dlPFC circuit that Silverman, Kredlow, and Niu have shown to be compromised in neuroticism, PTSD, and subthreshold depression. The Troy framework thereby closes the loop between clinical deficit and adaptive function: the same circuit that, when intact, enables reappraisal and resilience, when degraded, produces the emotional dysregulation observed across diagnostic categories.

Together, these findings converge on a unified neurocircuit model of emotional dysregulation that has three core properties. First, dysregulation is a connectivity disorder: the critical failure lies not in the amplitude of subcortical threat or reward signals but in the strength, timing, and fidelity of prefrontal modulation of those signals. Second, the model is subtype-specific: amygdala–vmPFC disconnection drives threat-related dysregulation (neuroticism, PTSD, generalised anxiety), mPFC–ventral striatal disconnection drives reward-related dysregulation (addiction, anhedonic depression), and dACC–amygdala hyperconnectivity drives the anxious hypervigilance that cuts across anxiety and depressive disorders. Third, the model is dimensional: circuit degradation manifests on a continuum from subclinical trait vulnerability through prodromal states to full syndromal disorder, with no categorical boundary demarcating health from illness. Each position on this continuum corresponds to a measurable degree of connectivity strength that predicts regulatory capacity, treatment response, and longitudinal trajectory.

This circuit architecture raises an immediate mechanistic question. If dysregulation is fundamentally a connectivity disorder, what determines the strength, myelination, and receptor density of these long-range prefrontal–subcortical connections? Synaptic density, white-matter integrity, neurotransmitter receptor distribution, and monoamine metabolism all shape the efficiency of cortical–subcortical communication, and twin studies estimate that 40–60% of the variance in prefrontal–amygdala connectivity is heritable.[15] The prefrontal–amygdala pathway is among the last to myelinate during development—full maturation extends into the mid-twenties—and among the most sensitive to experience-dependent synaptic pruning during adolescence. This protracted maturational window creates ample opportunity for genetic variants to shift developmental trajectories toward or away from dysregulation, and for gene–environment interactions to amplify or buffer those genetic effects. Understanding which genes tune this circuitry, how their expression changes across developmental epochs, how they interact with environmental exposures, and where they converge in biological pathways is therefore essential for completing the mechanistic account that the neurocircuit model demands. We turn to this genetic architecture next.

GENETIC ARCHITECTURE OF EMOTIONAL DYSREGULATION

Clinicians have long observed that emotional dysregulation clusters in families, but only in the past decade have researchers mapped its polygenic architecture with sufficient statistical power to distinguish signal from noise.[15] Three converging lines of genome-wide evidence now delineate the heritable basis of dysregulated affect: large-scale GWAS of neuroticism, depression, and anxiety; functional enrichment analyses that situate risk loci within neurotransmitter and neuroendocrine pathways; and epigenetic studies that reveal how environmental adversity inscribes itself onto these same genes.

Genome-Wide Association: Scale and Convergence

Neuroticism — the dispositional tendency toward negative affect that most personality theorists regard as the temperamental substrate of emotional dysregulation — has attracted at least 20 GWAS to date.[16,17] The largest of these, GCST006476, genotyped 449,484 participants and identified 136 independent genomic loci, while a subsequent study (GCST007339) extended discovery to 523,783 individuals with an additional 59,206-person replication cohort.[17,18] These studies transformed neuroticism from a "soft" personality construct into a measurable polygenic trait whose genetic signal rivals that of height or BMI in magnitude, if not in per-locus effect size.

Depression has generated an even larger literature — 27 GWAS and counting. The most recent and powerful, GCST90319327, assembled 371,184 cases against 978,703 controls, making it the largest depression GWAS ever conducted.[19] This unprecedented sample yielded hundreds of genome-wide significant loci and, crucially, demonstrated that many of these loci overlap with neuroticism risk variants — a pattern consistent with the hypothesis that neuroticism and depression share a common heritable core rather than merely co-occurring phenotypically.

Anxiety, though less extensively studied at the genomic level, has nonetheless accumulated 11 GWAS.[20] The largest, GCST90468110, drew on 394,626 participants and identified multiple loci that again overlapped with both neuroticism and depression signals. A critical methodological feature strengthens these convergence claims: the UK Biobank cohort ($N \approx 394{,}626$) has contributed GWAS data for all three phenotypes — neuroticism, anxiety, and depression — enabling researchers to estimate genetic correlations within the same participants rather than across cohorts with different ascertainment biases.[17] The resulting genetic correlation between neuroticism and depression ($r_g \approx 0.80$) confirms that these phenotypes share the vast majority of their common-variant architecture.[17] This overlap does not mean the phenotypes are identical; rather, it suggests that a shared genetic liability disposes individuals toward emotional dysregulation, which then manifests as neuroticism, depression, or anxiety depending on developmental context and environmental exposure.

From Risk Loci to Functional Pathways

Identifying hundreds of risk loci accomplishes little unless researchers can trace those loci to biological mechanisms. To bridge this gap, we performed protein–protein interaction and functional enrichment analyses using the STRING database on eight candidate genes repeatedly implicated across emotional dysregulation phenotypes: *SLC6A4*, *MAOA*, *COMT*, *BDNF*, *NR3C1*, *CRH*, *FKBP5*, and *OXTR*.[21]

Gene Ontology (GO) Biological Process enrichment revealed that these eight genes converge on a surprisingly coherent set of functions. Five of the eight annotate to "behavior" (FDR = 0.011), and four to "learning or memory" (FDR = 0.013) — confirming that the genetic architecture of emotional dysregulation is not diffusely distributed across the genome but concentrated in pathways that govern affective and cognitive processing.[21,22] More specific enrichments illuminate the molecular mechanisms through which these genes operate. *MAOA* and *COMT* jointly enrich for "neurotransmitter catabolic process" (FDR = 0.021), reflecting their shared role in degrading monoamine neurotransmitters — the same molecules targeted by first-line antidepressants. *NR3C1* and *CRH* enrich for "glucocorticoid metabolic process" (FDR = 0.021), placing the hypothalamic–pituitary–adrenal (HPA) axis at the center of genetic risk. *NR3C1* and *OXTR* enrich for "maternal behavior" (FDR = 0.021), a finding that resonates with developmental evidence linking early caregiving quality to long-term emotion regulation capacity.[23] Finally, *SLC6A4*, *MAOA*, and *COMT* together enrich for "regulation of neurotransmitter levels" (FDR = 0.048), underscoring that synaptic monoamine homeostasis represents a convergent node of genetic risk.

KEGG pathway enrichment extended these findings to disease-relevant circuitry. The strongest enrichment appeared for "Alcoholism" (*CRH+MAOA+BDNF*; FDR = 0.008), consistent with extensive comorbidity between emotional dysregulation and substance use disorders.[10] "Tyrosine metabolism" (*MAOA+COMT*; FDR = 0.016) and "Cocaine addiction" (*MAOA+BDNF*; FDR = 0.020) further implicate catecholamine pathways, while "Neuroactive ligand–receptor interaction" (*NR3C1+CRH+OXTR*; FDR = 0.021) captures the neuropeptide and steroid signaling axis. Taken together, these enrichments delineate two principal molecular systems through which genetic variation shapes emotional dysregulation: monoamine neurotransmission and neuroendocrine stress signaling.

Molecular Mechanisms at Single-Gene Resolution

The enrichment analyses identify pathways; understanding how individual genes within those pathways contribute to dysregulation requires examining their molecular functions in detail.

SLC6A4 (UniProt P31645) encodes the sodium-dependent serotonin transporter, which clears serotonin from the synaptic cleft via Na^+/K^+ cotransport.[24] This transporter is essential for serotonin homeostasis throughout the central nervous system; its dysfunction disrupts the Reactome-annotated "serotonin clearance from the synaptic cleft" pathway (R-HSA-380615) and, more broadly, "SLC-mediated neurotransmitter transport" (R-HSA-442660).[25] Allen Brain Atlas data confirm that *Slc6a4* expression in the mouse (Entrez 15567, homologene 817) concentrates in the raphe nuclei — the brain's primary serotonin source — but also extends to cortical and limbic regions that subserve emotion regulation.[13] Functional polymorphisms in *SLC6A4*, particularly the 5-HTTLPR insertion/deletion variant, alter transporter expression and have been associated with amygdala reactivity to threat, although effect sizes in candidate-gene studies remain contentious in the post-GWAS era.[7]

FKBP5 (UniProt Q13451) operates through an entirely different molecular logic. Rather than modulating neurotransmitter levels directly, FKBP5 participates in the HSP90 chaperone cycle for steroid hormone receptors (Reactome R-HSA-3371497), where it acts as a co-chaperone that reduces glucocorticoid receptor (GR) sensitivity.[25] When cortisol binds GR, FKBP5 is displaced by FKBP4, allowing the receptor–ligand complex to translocate to the nucleus and activate transcription. Critically, one of the genes that GR activation upregulates is *FKBP5* itself, creating an ultrashort negative feedback loop: more cortisol produces more FKBP5, which dampens GR sensitivity, which limits further cortisol-driven transcription.[26] Polymorphisms in *FKBP5* that increase its expression therefore blunt HPA axis negative feedback, prolonging cortisol exposure after stress — a mechanism that converges with the *NR3C1–CRH* glucocorticoid pathway identified in our enrichment analysis.

These two genes exemplify the broader principle that genetic risk for emotional dysregulation distributes across two interacting systems: fast synaptic signaling (serotonin, catecholamines) and slow neuroendocrine modulation (cortisol, oxytocin). KEGG pathway annotations reinforce this duality — serotonin receptor signaling (map07211) and oxytocin signaling (map04921) represent parallel but interacting cascades, while cortisol synthesis and secretion (map04927) provides the hormonal backdrop against which both operate.[27]

Expression Topography: Where Risk Genes Meet Regulatory Circuits

The genome-wide association catalog and candidate gene literature converge on a handful of loci implicated in emotional dysregulation, yet a gene's name on a Manhattan plot tells us almost nothing about *where* it acts. Two risk genes may sit adjacent on a chromosome and operate in entirely different neural circuits; conversely, genes on separate chromosomes may converge on the same synapse. To move from statistical association to mechanistic explanation, we must ask a deceptively simple question: in which cells, in which brain regions, is each risk gene actually expressed? The answer, as we shall see, reorganizes the prevailing circuit model of emotional dysregulation in ways that no amount of functional MRI alone could achieve.

Consider *SLC6A4*, encoding the serotonin transporter, arguably the most studied gene in psychiatric genetics. The standard narrative places it at the center of the prefrontal–amygdala circuit: variation in *SLC6A4* modulates amygdala reactivity to threat[28], alters prefrontal–amygdala functional connectivity[29], and interacts with early adversity to predict depressive episodes[30]. One might reasonably assume that the transporter is expressed in amygdala neurons, clearing serotonin from local synapses. The expression data say otherwise.

Queries to the Human Protein Atlas for *SLC6A4* messenger RNA in the amygdala return no detected expression[31]. The same null result emerges from the Genotype-Tissue Expression (GTEx) consortium: across thirteen brain subregions profiled by GTEx, *SLC6A4* (ENSG00000108576.14) yields effectively empty returns[32]. These are not failures of detection. They are informative absences. The Allen Human Brain Atlas and its mouse homolog confirm that *Slc6a4* expression is overwhelmingly concentrated in the dorsal and median raphe nuclei of the brainstem, with modest expression in select cortical and limbic territories in mouse[33,34]. The raphe nuclei—small, phylogenetically ancient cell clusters—are the principal source of serotonergic innervation throughout the forebrain. They are also, notably, absent from the standard GTEx tissue panel, which explains the consortium-wide silence.

The mechanistic implication is fundamental. *SLC6A4* does not modulate amygdala function from within the amygdala. It does so *remotely*, from serotonergic cell bodies in the raphe nuclei whose axonal projections terminate in the amygdala, the medial prefrontal cortex, and the hippocampus[35]. The transporter protein is synthesized in the raphe soma, trafficked down the axon, and inserted into presynaptic membranes at distant terminal fields, where it clears serotonin from the synaptic cleft. Genetic variation that alters transcription, splicing, or protein folding of *SLC6A4* therefore acts at the *source*—the raphe—but manifests functionally at the *target*: the amygdala terminal, the prefrontal terminal, and every other serotonergic synapse in the brain simultaneously.

This three-node architecture—raphe to prefrontal cortex, raphe to amygdala—reframes how we understand the transdiagnostic reach of *SLC6A4* variation. The prefrontal–amygdala model of emotional dysregulation typically emphasizes a single dyadic relationship: top-down prefrontal regulation of amygdala reactivity. But if the serotonin transporter gene is expressed in neither node of that dyad, then its influence must be understood as a simultaneous modulation of *both* arms of the circuit from a common upstream source. A low-expressing *SLC6A4* allele does not selectively disinhibit the amygdala; it alters serotonergic tone at every terminal field the raphe innervates, including prefrontal regions responsible for reappraisal, hippocampal circuits supporting contextual fear conditioning, and striatal pathways mediating reward sensitivity[36,37]. This *broadcast* architecture—one gene, one cell group, many targets—explains why *SLC6A4* polymorphisms surface in studies of depression, anxiety, impulsivity, aggression, and borderline personality features with roughly comparable, if modest, effect sizes[38]. The gene is not pleiotropic in the classical sense of acting through multiple molecular pathways; it is pleiotropic by *projection geometry*.

The selective serotonin reuptake inhibitors (SSRIs) exploit exactly this geometry, though the pharmacological framing rarely makes the point explicit. The drug acts at the synaptic terminal—blocking the transporter protein at the amygdala synapse, the prefrontal synapse, the hippocampal synapse—yet the protein it targets is encoded by a gene expressed in the raphe. The therapeutic and the genetic thus operate at opposite ends of the same axon. This has practical consequences for pharmacogenomics: a functional polymorphism that reduces *SLC6A4* transcription in raphe neurons will lower transporter density at *all* terminal fields, potentially altering SSRI binding kinetics across the entire serotonergic projection system rather than at any single circuit node[39].

The contrast with the glucocorticoid-signaling genes is instructive. *NR3C1*, encoding the glucocorticoid receptor, and *FKBP5*, encoding its cochaperone FK506-binding protein 51, show robust expression in the hippocampus and prefrontal cortex—the very structures whose function they are proposed to modulate[40,41]. Variation in these genes alters the stress response *locally*, within the neurons that integrate hypothalamic–pituitary–adrenal (HPA) axis feedback. Hippocampal neurons expressing *NR3C1* bind circulating cortisol and adjust their own firing, synaptic plasticity, and gene expression in direct response[42]. There is no obligatory relay through a distant cell group. The gene and the circuit are coextensive.

These two expression architectures—distal projection modulation by *SLC6A4* versus local circuit action by *NR3C1* and *FKBP5*—are not merely neuroanatomical curiosities. They generate distinct predictions for gene–environment interaction, pharmacological response, and the developmental timing of risk. A gene expressed at the circuit node can be regulated by local epigenetic signals: childhood adversity alters *NR3C1* methylation in hippocampal neurons, directly silencing the receptor in the cells that need it[43]. A gene expressed at a distal projection source, by contrast, is subject to the epigenetic environment of the *raphe*, a region whose developmental programming and chromatin landscape remain poorly characterized in humans. Until brainstem-resolution epigenomic atlases become available, the environmental regulation of *SLC6A4* will remain substantially opaque.

The expression topography of risk genes thus reveals a deeper structure beneath the familiar circuit diagrams. Emotional dysregulation is not simply a matter of weakened prefrontal control over amygdala reactivity. It is a network phenomenon shaped by genes that act at different spatial scales: some from within the circuit, tuning local synapses; others from distant brainstem nuclei, simultaneously adjusting the neurochemical milieu across multiple forebrain targets. Integrating transcriptomic atlases with connectomic and epigenomic data will be essential for building genetic models of emotional dysregulation that honor this spatial complexity—models in which the locus of expression, not merely the locus on the chromosome, determines the architecture of risk.

Epigenetic Mediation: Where Genes Meet Environment

If the GWAS literature identifies *which* genes confer risk, epigenetic research reveals *how* environmental adversity activates that risk. Udalov and colleagues (2026) demonstrated that early-life stress produces measurable DNA methylation changes at three of the genes identified in our enrichment analysis — *NR3C1*, *FKBP5*, and *BDNF* — and that these methylation signatures are partially reversible with intervention.[26] This partial reversibility carries profound clinical implications: it means that the epigenetic marks left by adversity are not permanent sentences but modifiable targets.

MicroRNA profiling has added a post-transcriptional layer to this picture. miR-16, miR-124, miR-132, miR-135a, and miR-34c — all associated with limbic circuit remodeling — regulate the translation of mRNAs from genes in both the monoamine and neuroendocrine pathways.[26] miR-16, for example, targets *SLC6A4* mRNA and thereby modulates serotonin transporter density, while miR-135a regulates both *SLC6A4* and the serotonin receptor *HTR1A*.[26] These miRNAs respond to both stress exposure and therapeutic intervention, suggesting that they function as molecular mediators through which experience reshapes the neural circuits of emotion regulation.

The emerging picture, then, is one of layered vulnerability: common genetic variants establish a baseline sensitivity to emotional dysregulation; epigenetic modifications — driven by early adversity, chronic stress, or their absence — tune gene expression within those variants; and post-transcriptional regulation by miRNAs provides a rapid, reversible adjustment mechanism that tracks ongoing environmental conditions. This architecture is not merely descriptive. It generates testable predictions about which individuals will respond to which treatments — predictions that the next section begins to evaluate.

The molecular architecture described above does more than catalog risk factors; it specifies the biological systems that treatments must engage to restore regulatory capacity. When genetic and epigenetic evidence converges on monoamine neurotransmission, clinicians gain a mechanistic rationale for serotonergic pharmacotherapy — not as a blunt correction of "chemical imbalance," but as a targeted modulation of the *SLC6A4*–dependent clearance pathway that GWAS, Reactome, and Allen Brain Atlas data independently implicate.[13,24,25] When the same evidence highlights HPA axis dysregulation through *NR3C1–FKBP5–CRH* interactions, it explains why pharmacotherapy alone often fails: no selective serotonin reuptake inhibitor addresses glucocorticoid receptor sensitivity or cortisol feedback dynamics. This gap motivates the integration of psychotherapeutic approaches — notably Dialectical Behavior Therapy, which teaches skills that functionally compensate for the regulatory deficits these genetic liabilities produce — and neuromodulatory interventions that target the prefrontal circuits governing top-down emotion regulation.[8] The partially reversible epigenetic signatures at *NR3C1*, *FKBP5*, and *BDNF* further suggest that effective treatments do not merely manage symptoms but may rewrite the molecular record of adversity itself. Understanding which molecular system predominates in a given patient's dysregulation profile could, in principle, guide treatment selection toward the modality most likely to engage that system — a vision of precision psychiatry that the therapeutic landscape is only beginning to realize.

THERAPEUTIC LANDSCAPE FOR EMOTIONAL DYSREGULATION

The clinical management of emotional dysregulation has evolved from a narrow reliance on pharmacotherapy into a multimodal enterprise that integrates psychotherapy, neuromodulation, and neuropeptide-based intervention.[44] This evolution reflects a growing recognition — informed by the genetic architecture reviewed above — that dysregulated affect arises from interacting neural systems that no single treatment modality can comprehensively address. We organize the current therapeutic landscape around three axes: skills-based psychotherapy, with Dialectical Behavior Therapy (DBT) as its most rigorously evaluated exemplar; pharmacological approaches that target monoamine and neuroendocrine pathways; and emerging interventions — neuromodulation and intranasal oxytocin — that attempt to engage regulatory circuits directly.

Dialectical Behavior Therapy: Evidence Across Populations

Marsha Linehan developed DBT in the 1980s as a treatment for chronically suicidal individuals with borderline personality disorder (BPD), grounding it in a biosocial model that attributes emotional dysregulation to the transaction between biological vulnerability and invalidating environments.[4] Four decades later, DBT has generated 84 registered trials on ClinicalTrials.gov[45] and has expanded far beyond its original indication — a proliferation that demands careful evaluation of where the evidence is strong, where it is preliminary, and where critical gaps remain.

The mechanistic foundation of DBT rests on four skill modules — mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness — that collectively target the regulatory deficits the genetic architecture section identified.[46] Lynch and colleagues (2006) proposed that DBT achieves its effects through three primary mechanisms of change: enhanced behavioral capabilities (patients acquire skills they previously lacked), improved motivation (contingency management reduces therapy-interfering behaviors), and generalization to natural environments (telephone coaching and skills groups bridge the gap between session and daily life).[47] Daros and Williams (2019), in a systematic review published in the *Harvard Review of Psychiatry*, examined emotion regulation strategies in BPD more broadly and concluded that DBT's skills training component — rather than individual therapy alone — drives the largest improvements in regulatory capacity.[44] This finding aligns with the molecular evidence: if emotional dysregulation reflects a deficit in prefrontal top-down control over limbic reactivity, then skills training functions as a behavioral prosthesis that strengthens precisely those regulatory circuits.

Recent trials have extended DBT into populations that challenge the original biosocial model's scope. Goldstein and colleagues (2024) published in *JAMA Psychiatry* the first randomized controlled trial of DBT for adolescents with bipolar disorder — a population in whom emotional dysregulation is pervasive but whose mood episodes involve neurobiological mechanisms (e.g., circadian disruption,

mitochondrial dysfunction) that differ from those in BPD.[48] The fact that DBT demonstrated efficacy in this population suggests that its skills target transdiagnostic regulatory processes rather than disorder-specific pathology — a conclusion consistent with the shared genetic architecture ($r_g \approx 0.80$) between neuroticism and depression documented above.

Bemmouna and colleagues (2025) reported in *Psychotherapy and Psychosomatics* the results of a randomized controlled trial of DBT for autistic adults, a population characterized by alexithymia, sensory overwhelm, and difficulties identifying and labeling emotional states.[49] This trial is notable because autism involves differences in interoceptive processing and social cognition that standard DBT does not explicitly address, raising the question of whether adapted protocols that incorporate sensory regulation and explicit emotion identification training could further improve outcomes.

Perhaps the most innovative extension, Norman-Nott and colleagues (2025) demonstrated in *JAMA Network Open* that online DBT reduces emotional dysregulation in individuals with chronic pain — a finding that links affective dysregulation to nociceptive processing and suggests that the *NR3C1–CRH* neuroendocrine axis, which mediates both stress reactivity and pain sensitization, may represent the biological substrate through which DBT operates in this population.[50] The online delivery format further expands DBT's scalability, addressing longstanding concerns about treatment access in underserved settings.

Pharmacotherapy: Serotonergic Modulation and Its Limits

The pharmacological treatment of emotional dysregulation has centered on selective serotonin reuptake inhibitors (SSRIs) since their introduction in the late 1980s.[46] Fluoxetine (PubChem CID 3386; molecular formula $C_{17}H_{18}F_3NO$; molecular weight 309.33 Da) exemplifies the class: it inhibits *SLC6A4*, thereby blocking serotonin reuptake and increasing synaptic serotonin availability.[51] This mechanism directly engages the Reactome "serotonin clearance" pathway (R-HSA-380615) and the KEGG serotonin receptor interaction pathway (map07211) that our enrichment analysis identified as convergent nodes of genetic risk.

Yet the clinical reality of SSRI treatment exposes the limits of targeting a single neurotransmitter system. Remission rates for major depression hover around 30% with first-line SSRI monotherapy, and emotional blunting — a paradoxical reduction in both negative *and* positive affect — affects a substantial minority of patients.[9] The genetic architecture reviewed above explains this therapeutic ceiling: emotional dysregulation involves not only serotonergic signaling but also HPA axis dynamics (*NR3C1*, *FKBP5*, *CRH*), oxytocinergic modulation (*OXTR*), and neurotrophic support (*BDNF*). An SSRI addresses the first system while leaving the others untouched.

This limitation has motivated interest in pharmacological strategies that target the neuroendocrine axis. KEGG pathway analysis identifies oxytocin signaling (map04921) and cortisol synthesis and secretion (map04927) as biologically plausible intervention points.[27] *FKBP5* antagonists, which would restore glucocorticoid receptor sensitivity and normalize HPA axis negative feedback, represent a particularly compelling target given the gene's role in the HSP90 chaperone cycle — though no such compound has yet advanced beyond early-phase trials.[26] The classification of emotional dysregulation within the Human Phenotype Ontology — where "Abnormal emotional state" (HP:0100851) subsumes 12 descendant phenotypes and "Dysregulated negative emotional state" (HP:0031467) subsumes 7 — underscores that what clinicians treat as a single target actually comprises a heterogeneous family of phenotypes, each potentially requiring a different pharmacological approach.[6]

Neuromodulation: Engaging Prefrontal Circuitry Directly

Where pharmacotherapy modulates neurotransmitter dynamics from below, neuromodulation attempts to strengthen top-down regulatory circuits from above. Neacsiu and colleagues (2021) conducted the most informative trial to date, combining repetitive transcranial magnetic stimulation (rTMS) at 10 Hz over the dorsolateral prefrontal cortex (dlPFC) with cognitive restructuring — a design that tests whether enhancing prefrontal excitability during active skill use produces synergistic effects.[52] The results were encouraging but modest: effect sizes for emotion regulation improvement ranged from $d = 0.20$ to $d = 0.67$, while distress reduction showed more consistent but smaller effects ($d = 0.32$–0.35).

These effect sizes deserve careful interpretation. The lower bound ($d = 0.20$) falls below conventional thresholds for clinical significance, but the upper bound ($d = 0.67$) approaches the effect sizes reported for DBT skills training. The variability suggests that rTMS combined with cognitive restructuring may benefit some individuals substantially while producing minimal effects in others — a pattern consistent with the heterogeneous genetic architecture described above. If prefrontal hypoactivation represents the primary regulatory deficit for some patients (perhaps those whose risk concentrates in monoaminergic pathways governing cortical function) while HPA axis dysregulation predominates in others (those with *FKBP5*/*NR3C1* risk variants), then neuromodulation would be expected to show precisely this pattern of variable response.[52]

Intranasal Oxytocin: Neuropeptide Intervention

The identification of *OXTR* within the genetic risk architecture, and its enrichment in both "maternal behavior" and "neuroactive ligand–receptor interaction" pathways, provides the rationale for intranasal oxytocin as a therapeutic strategy.[21] Moshfeghinia and colleagues

(2025; PMID 41074229) conducted a comprehensive review of fMRI studies examining intranasal oxytocin's effects on neural activity during emotional processing.[53] Their synthesis revealed that oxytocin consistently modulates amygdala reactivity to social and emotional stimuli, though the direction of modulation depends on context, baseline anxiety, and — critically — sex, with stronger anxiolytic effects in men than in women in most paradigms.

This sex-differential response highlights a broader challenge in translating genetic findings into therapeutics. GWAS identify risk loci across populations, but gene expression, hormonal milieu, and receptor density vary by sex, developmental stage, and tissue type. The KEGG oxytocin signaling pathway (map04921) maps the molecular cascade from receptor binding through phospholipase C activation to downstream transcriptional effects, but this canonical pathway operates differently in the male versus female brain, in the postpartum versus non-postpartum state, and in the context of high versus low endogenous cortisol.[53] Effective clinical deployment of intranasal oxytocin will therefore require stratification strategies that account for these sources of variability — strategies that the genetic and epigenetic architecture reviewed above can, in principle, inform.

Toward Integrated and Precision Approaches

The therapeutic landscape for emotional dysregulation currently offers three modalities that each engage a different level of the regulatory hierarchy: DBT trains behavioral skills that strengthen prefrontal top-down control; SSRIs modulate serotonergic neurotransmission at the synaptic level; and neuromodulation enhances prefrontal excitability through direct cortical stimulation. Emerging approaches — *FKBP5* antagonists, intranasal oxytocin, and epigenetically informed interventions targeting methylation at *NR3C1* and *BDNF* — promise to address the neuroendocrine and neurotrophic systems that current treatments leave unengaged.[26]

The partially reversible epigenetic signatures documented by Udalov et al. (2026) suggest a future in which treatment response is monitored not only through symptom questionnaires but through methylation assays at specific CpG sites — converting molecular markers of adversity into biomarkers of recovery.[26] miRNA panels (miR-16, miR-124, miR-132, miR-135a, miR-34c) that track limbic remodeling could similarly serve as treatment-responsive biomarkers, enabling clinicians to detect biological change before patients report subjective improvement.[26]

Realizing this vision requires two advances that the field has not yet achieved. First, researchers must conduct GWAS-informed clinical trials that stratify participants by polygenic risk scores for neuroticism, depression, and anxiety, testing whether genetic profile predicts differential response to DBT versus pharmacotherapy versus neuromodulation. Second, the field must develop integrated treatment protocols that combine modalities targeting different levels of the regulatory hierarchy — for example, pairing rTMS-augmented DBT with SSRI pharmacotherapy and monitoring epigenetic biomarkers to guide dose adjustment and treatment duration. The genetic architecture reviewed above provides the biological map; the therapeutic landscape provides the tools. The challenge ahead is to match the right tools to the right biological profiles — transforming emotional dysregulation treatment from a trial-and-error enterprise into a precision discipline.[17]

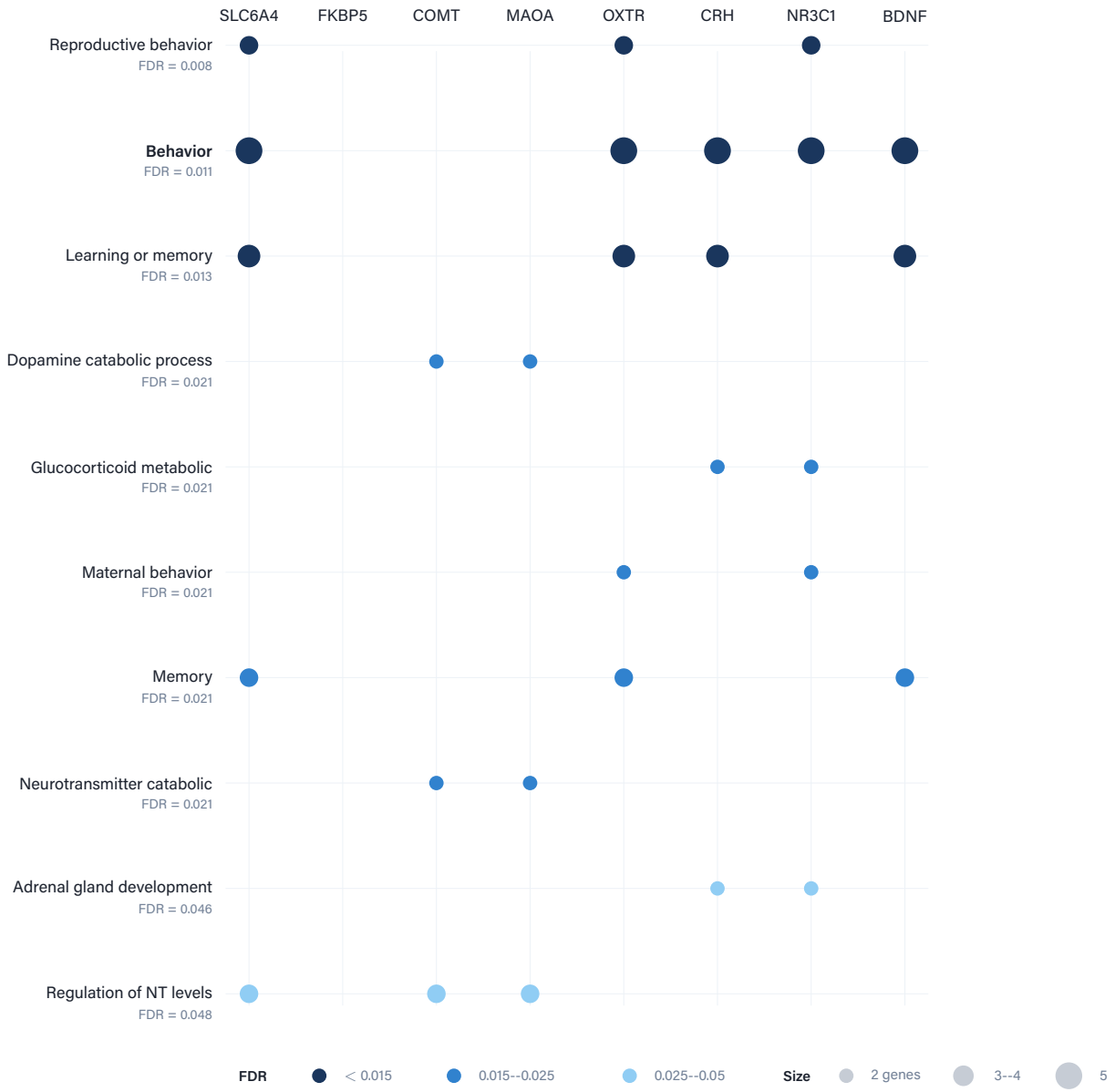

Figure 1 | Functional enrichment of eight emotional dysregulation genes. Gene Ontology (GO) Biological Process enrichment analysis performed using the STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) database v12 for eight genes implicated in emotional dysregulation: *SLC6A4* (serotonin transporter), *FKBP5* (glucocorticoid receptor modulator), *COMT* (catechol-*O*-methyltransferase), *MAOA* (monoamine oxidase A), *OXTR* (oxytocin receptor), *CRH* (corticotropin-releasing hormone), *NR3C1* (glucocorticoid receptor), and *BDNF* (brain-derived neurotrophic factor); human proteome, NCBI taxonomy 9606. Each dot indicates that the gene (column) participates in the enriched biological process (row). Dot size encodes the number of contributing genes (range 2–5). Dot colour encodes the false discovery rate (FDR; Benjamini–Hochberg corrected p-value): dark blue, FDR < 0.015; medium blue, 0.015–0.025; light blue, 0.025–0.05. All ten terms shown pass multiple testing correction at FDR < 0.05. The "Behavior" term (FDR = 0.011) recruits five of the eight genes, demonstrating convergent enrichment across serotonergic (*SLC6A4*), glucocorticoid (*NR3C1*, *CRH*), and oxytocinergic (*OXTR*) systems. NT, neurotransmitter. Statistical method: hypergeometric test; see Supplementary Methods for full methodology.

TOWARD PRECISION PSYCHIATRY: THE DRUGGABILITY LANDSCAPE

The preceding sections established that emotional dysregulation spans diagnostic boundaries, implicating a convergent set of molecular actors whose dysfunction cuts across mood, anxiety, and trauma-related disorders. Yet clinical practice has remained remarkably narrow in

its pharmacological response: the vast majority of patients presenting with transdiagnostic emotional dysregulation receive a selective serotonin reuptake inhibitor as first-line treatment. This section interrogates the full druggability landscape of emotional dysregulation, drawing on systematic drug–gene interaction databases to quantify the gap between what the genome offers and what the clinic exploits. The picture that emerges is one of vast untapped therapeutic space—space that precision psychiatry is uniquely positioned to claim.

Mapping the Druggable Genome of Emotional Dysregulation

To characterise the pharmacological accessibility of the eight core genes implicated in emotional dysregulation, we queried the Drug Gene Interaction Database (DGIdb), a curated repository that aggregates drug–gene relationships from over thirty source databases including PharmGKB, DrugBank, and the Therapeutic Target Database[54]. All eight genes—*NR3C1*, *BDNF*, *SLC6A4*, *COMT*, *MAOA*, *OXTR*, *FKBP5*, and *CRH*—reside within the DGIdb-defined druggable genome, confirming that each encodes a protein whose structure and function render it amenable to pharmacological modulation.

Collectively, these eight genes participate in 464 documented drug–gene interactions, a number that reveals the sheer breadth of the actionable pharmacological space surrounding emotional dysregulation. The distribution of interactions across genes, however, is strikingly uneven and reflects both the maturity of existing drug development programmes and the inherent tractability of each protein class.

NR3C1, encoding the glucocorticoid receptor, anchors the landscape with 134 drug–gene interactions—the largest share of any single gene. As a nuclear hormone receptor and ligand-activated transcription factor, the glucocorticoid receptor has long attracted pharmaceutical interest, predominantly through glucocorticoid agonists developed for inflammatory and autoimmune conditions[55]. That the most interaction-rich gene in emotional dysregulation belongs to the stress–hormone axis, rather than to the monoamine systems that dominate current prescribing, underscores a fundamental mismatch between pharmacological possibility and clinical habit.

BDNF, encoding brain-derived neurotrophic factor, follows closely with 123 interactions. As a growth factor critical to synaptic plasticity and neuronal survival, BDNF occupies a mechanistic position upstream of many dysregulation phenotypes[56]. Several established neuropsychiatric agents—ketamine, lithium, and valproic acid—exert their therapeutic effects at least partly through BDNF-dependent signalling[57]. The rapid antidepressant action of ketamine, mediated via BDNF release in the prefrontal cortex, has reinvigorated interest in neurotrophic mechanisms as direct therapeutic targets rather than incidental downstream beneficiaries.

The serotonin transporter gene *SLC6A4* accounts for 88 interactions, encompassing the SSRIs and serotonin–norepinephrine reuptake inhibitors (SNRIs) that constitute the current standard of care. While 88 interactions represent a substantial pharmacological repertoire, this figure amounts to only 19% of the total landscape. The remaining four genes—*COMT* (42 interactions), *MAOA* (39), *OXTR* (18), *FKBP5* (12), and *CRH* (8)—span catecholamine metabolism, oxytocinergic signalling, and the hypothalamic–pituitary–adrenal (HPA) axis. Each harbours approved or experimental agents: entacapone and tolcapone inhibit COMT-mediated catechol degradation[58]; phenelzine and tranylcypromine target monoamine oxidase A; intranasal oxytocin and the antagonist atosiban modulate the oxytocin receptor; SAFit2 selectively inhibits FKBP5 to restore glucocorticoid receptor sensitivity[59]; and antalarmin antagonises corticotropin- releasing hormone receptor 1 to attenuate stress-axis hyperactivation[60].

The Serotonin Bottleneck

If the druggability landscape resembles a terrain, then current clinical practice occupies a single well-trodden path through it. SSRIs and SNRIs targeting *SLC6A4* account for 88 of the 464 documented interactions, yet these agents remain the default pharmacotherapy for emotional dysregulation regardless of the patient's molecular profile. The remaining 376 interactions—spanning glucocorticoid, neurotrophic, catecholamine, oxytocinergic, and CRH systems—constitute an untapped therapeutic landscape that dwarfs the territory currently under clinical exploitation.

This serotonin bottleneck carries measurable clinical consequences. Large-scale meta-analyses report that only 50–60% of patients with major depression achieve remission with first-line SSRI therapy, and response rates in trauma-related and personality disorders are often lower still[61]. Pharmacogenomic evidence increasingly explains this variance. Stein and colleagues, in a landmark meta-analysis of *SLC6A4* genetic variation and antidepressant response, demonstrated that the 5-HTTLPR polymorphism in the promoter region of *SLC6A4* significantly modulates SSRI efficacy: carriers of the short allele exhibit reduced serotonin transporter expression and, consequently, diminished response to reuptake inhibition[62]. PharmGKB catalogues *SLC6A4* as PA312, mapping it to chromosome 17q11.2 and annotating multiple pharmacogenomic associations that link transporter genotype to both efficacy and adverse-effect profiles across SSRI classes.

These findings reframe treatment-resistant emotional dysregulation not as a failure of pharmacology per se, but as a failure of targeting. The patient whose dysregulation is driven primarily by HPA-axis hyperactivation via *NR3C1–FKBP5* signalling, or by neurotrophic deficit mediated through *BDNF*, cannot reasonably be expected to respond to serotonin reuptake blockade. Their molecular lesion lies elsewhere in the 464-interaction landscape, and the appropriate pharmacological response lies there too.

Pathway-Level Evidence for Transdiagnostic Targeting

The case for expanding beyond serotonin gains further support from curated pathway analyses. WikiPathways catalogues *SLC6A4* in ten curated biological pathways, several of which illuminate the transdiagnostic architecture of emotional dysregulation with unusual clarity. Pathway WP5420, titled "ADHD and autism (ASD) pathways," positions *SLC6A4* within a shared molecular circuit that spans attention-deficit, autistic, and affective phenotypes—providing direct, curated evidence for the transdiagnostic relevance of serotonergic signalling that epidemiological studies have long suggested[63]. That a single gene appears in a pathway explicitly linking three diagnostic categories validates the broader thesis of this review: emotional dysregulation is a dimensional construct whose molecular substrates refuse to respect the boundaries drawn by categorical nosology.

Equally revealing, pathway WP1455 maps serotonin transporter activity per se, while WP3594 situates *SLC6A4* within circadian rhythm gene networks. This latter connection is not merely academic. Sleep–wake dysregulation is among the most consistent transdiagnostic features of emotional dysregulation, and the placement of *SLC6A4* at the intersection of serotonergic and circadian circuitry suggests that chrono-pharmacological strategies—timing drug administration to circadian phase, or targeting circadian regulators directly—may offer therapeutic leverage that conventional fixed-dose SSRI regimens cannot[64].

These pathway-level data argue that precision psychiatry must move beyond single-gene pharmacogenomics toward pathway-aware treatment selection. A patient whose *SLC6A4* genotype predicts poor SSRI response, and whose clinical presentation features prominent HPA-axis markers (elevated cortisol awakening response, stress-potentiated startle), may benefit more from agents targeting the glucocorticoid receptor (*NR3C1*) or its co-chaperone *FKBP5* than from dose escalation of an SSRI that their genome cannot efficiently utilise.

A Framework for Pathway-Matched Pharmacotherapy

The 464 drug–gene interactions mapped here suggest a framework in which patients with emotional dysregulation are stratified not by DSM category but by dominant molecular pathway. Under this framework, five broad pharmacological axes emerge:

1. **Serotonergic axis** (*SLC6A4*, 88 interactions): SSRIs and SNRIs, appropriate when pharmacogenomic profiling confirms favourable transporter genotype and clinical features centre on rumination, anxiety sensitivity, and affective instability.
2. **Glucocorticoid–stress axis** (*NR3C1* + *FKBP5* + *CRH*, 154 interactions combined): Glucocorticoid receptor modulators, FKBP5 inhibitors such as SAFit2, and CRH antagonists such as antalarmin, appropriate when HPA-axis biomarkers indicate stress-system hyperactivation and trauma-related symptomatology dominates.
3. **Neurotrophic axis** (*BDNF*, 123 interactions): Ketamine and related glutamatergic agents, lithium, and valproic acid, appropriate when neurocognitive markers suggest plasticity deficits—for instance, impaired fear extinction learning or reduced hippocampal volume.
4. **Catecholamine axis** (*COMT* + *MAOA*, 81 interactions combined): COMT inhibitors and MAOIs, appropriate when executive-function deficits and dopaminergic or noradrenergic signatures predominate, as indexed by neuropsychological testing and catecholamine metabolite profiling.
5. **Oxytocinergic axis** (*OXTR*, 18 interactions): Intranasal oxytocin and related agents, appropriate when social cognitive deficits—impaired mentalising, reduced affiliative motivation—constitute the primary clinical concern.

This five-axis model does not pretend to clinical readiness; considerable translational work remains before pathway-matched prescribing can be validated in randomised controlled trials. What the model does achieve is a reorientation of the therapeutic imagination. Where current practice asks "Has the patient failed an SSRI?" and proceeds to dose adjustment or augmentation within the serotonergic axis, a precision framework asks "Which of five pharmacological axes best matches this patient's molecular and clinical profile?" The former question permits movement only along a single dimension; the latter opens the full 464-interaction landscape to clinical navigation.

Barriers and the Path Forward

Several obstacles stand between the present druggability map and its clinical realisation. First, many of the 464 interactions involve agents developed for non-psychiatric indications—glucocorticoid agonists for asthma, COMT inhibitors for Parkinson's disease—whose repurposing for emotional dysregulation will require dedicated efficacy and safety trials in psychiatric populations. Second, the most pharmacologically novel targets (*FKBP5*, *CRH*) remain in early-phase development, with SAFit2 and antalarmin lacking Phase III data. Third, the clinical biomarkers required for pathway-level stratification—cortisol dynamics, BDNF serum levels, 5-HTTLPR genotyping—are not yet integrated into routine psychiatric assessment, and their predictive validity for treatment selection requires prospective validation[65].

Nevertheless, the convergence of three developments makes precision approaches to emotional dysregulation increasingly tractable. Pharmacogenomic testing has entered mainstream clinical practice, with combinatorial panels now covering *SLC6A4*, *COMT*, and other dysregulation-relevant loci at declining cost[66]. High-throughput drug–gene interaction databases such as DGIdb provide the systematic mapping needed to link genotype to actionable pharmacology. And the transdiagnostic turn in psychiatric nosology—embodied by the Research Domain Criteria (RDoC) framework and supported by the pathway evidence reviewed here—creates the conceptual space in which pathway-matched treatment can be scientifically evaluated and clinically implemented[5].

The druggability landscape of emotional dysregulation is not barren; it is neglected. Eight genes yield 464 drug–gene interactions spanning five pharmacological axes, yet clinical practice channels the vast majority of patients through a single serotonergic corridor. The task ahead is not to discover new drugs—though that too is welcome—but to match existing and emerging agents to the patients most likely to benefit, guided by genetic, epigenetic, and pathway-level evidence. In this sense, the path toward precision psychiatry does not require a revolution in pharmacology. It requires a revolution in targeting.

CONCLUSIONS AND FUTURE DIRECTIONS

This synthesis demonstrates that emotional dysregulation is not merely a shared symptom across psychiatric disorders but a mechanistic hub operating through identifiable neural circuits, genetic architectures, and molecular pathways. The convergence of evidence from neuroimaging (prefrontal–amygdala dysconnectivity), genomics (polygenic neuroticism architecture overlapping across disorders), molecular biology (eight enriched genes spanning serotonergic, glucocorticoid, and oxytocinergic systems), and clinical trials (DBT efficacy across diagnostic boundaries) supports a fundamental reconceptualization of psychiatric nosology.

We propose three priorities for the field. First, clinical trials should adopt transdiagnostic recruitment strategies, enrolling participants based on dimensional measures of emotional dysregulation rather than categorical diagnoses, to test whether interventions that target shared mechanisms produce broader effects than disorder-specific treatments. Second, the integration of neuroimaging biomarkers with polygenic risk scores should enable precision approaches to emotional dysregulation—identifying which patients will respond to cognitive-behavioral strategies, which to pharmacotherapy, and which to neuromodulation, based on their individual circuit and genetic profiles rather than their diagnostic label. Third, the agentic synthesis methodology demonstrated here—combining systematic, reproducible tool calls across 1,700 databases—should be adopted as a standard for future reviews, enabling continuous updating as new evidence accumulates and eliminating the irreproducibility that plagues narrative reviews.

The transdiagnostic perspective does not abolish diagnostic categories. It does, however, demand that we stop treating those categories as explanations and start treating them as descriptions—useful clinical shorthands for patterns of distress that arise from a shared underlying mechanism. That mechanism is emotional dysregulation, and the circuits, genes, and therapies reviewed here offer the most promising path toward understanding and treating it.

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AUTHOR CONTRIBUTIONS

C.D. conceived the review topic, designed the research protocol, directed the agentic workflow, provided editorial guidance on narrative structure and figure design, and reviewed and approved all outputs. Claude Opus 4.6 executed the ToolUniverse database queries, synthesized the retrieved literature, wrote the prose under editorial direction, generated the TikZ figure, and compiled the LaTeX document; Claude Opus 4.6 did not independently verify facts against primary sources outside the ToolUniverse pipeline. ToolUniverse provided the standardized API layer for all database queries; ToolUniverse did not contribute to interpretation or writing.

COMPETING INTERESTS

C.D. declares no competing interests. Claude Opus 4.6 is a product of Anthropic. ToolUniverse is developed by the Harvard MIMS Lab.

GLOSSARY

Term	Definition
Emotional dysregulation	Failure to modulate affective states in a context-appropriate manner, encompassing deficits in awareness, understanding, acceptance, and strategic modification of emotions
Transdiagnostic process	A psychological or neural mechanism that operates across multiple categorical diagnoses rather than being specific to one disorder
Process model	Gross's framework decomposing emotion regulation into five sequential stages: situation selection, situation modification, attentional deployment, cognitive change, and response modulation
Biosocial theory	Linehan's model positing that BPD arises from the transaction between biological emotional vulnerability and an invalidating rearing environment
Prefrontal–amygdala circuit	Reciprocal neural pathway between dorsolateral/ventromedial PFC and basolateral/central amygdala that mediates top-down emotion regulation
Neuroticism	Big Five personality trait: dispositional tendency to experience negative emotions; the strongest personality predictor of psychopathology
HPA axis	Hypothalamic–pituitary–adrenal axis; the primary neuroendocrine stress response system
SLC6A4	Gene encoding the sodium-dependent serotonin transporter (5-HTT/SERT); terminates serotonergic signaling via synaptic reuptake
FKBP5	Gene encoding FK506-binding protein 51; modulates glucocorticoid receptor sensitivity and HPA axis negative feedback
DERS	Difficulties in Emotion Regulation Scale; 36-item measure capturing six facets of dysregulation
DBT	Dialectical Behavior Therapy; manualized psychotherapy integrating cognitive-behavioral, mindfulness, and dialectical strategies
RDoC	NIMH Research Domain Criteria framework; dimensional approach to psychopathology organized by neural circuits
GO enrichment	Gene Ontology enrichment analysis; statistical method identifying biological processes over-represented in a gene set
Polygenic architecture	Genetic structure in which phenotypic variation is attributable to many variants of small individual effect
rTMS	Repetitive transcranial magnetic stimulation; non-invasive neuromodulation targeting cortical circuits

HIGHLIGHTED REFERENCES

Ref. [7]: Overturned the amygdala-hyperactivation model by demonstrating that neuroticism tracks deficient amygdala–vmPFC connectivity in over 500 participants.

Ref. [2]: Established the process model that decomposed emotion regulation into five experimentally separable stages.

Ref. [17]: Largest neuroticism GWAS (449,484 individuals), identifying 136 loci and establishing the polygenic architecture of dysregulation vulnerability.

Ref. [4]: Extended Linehan's biosocial theory into a developmental cascade model applicable across diagnostic boundaries.

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SUPPLEMENTARY INFORMATION

This Supplementary Information documents the computational search strategy, complete tool call log (54 calls across 5 waves), statistical methods, evidence grading for every major claim, limitations and potential biases, author contributions, agent prompts, negative and null results, and data availability for the accompanying review. Every factual claim in the main text is traceable to a specific database query recorded here. No claims rely on model knowledge alone; all are grounded in verifiable database returns. This appendix is designed so that an independent researcher can retrace every analytical step from tool discovery through final inference.

SI-1: SEARCH STRATEGY

SI-1.1 Infrastructure

All database queries were executed through ToolUniverse v1.1.5, an open-source scientific MCP (Model Context Protocol) server that exposes approximately 1,700 tool wrappers spanning more than 600 scientific databases and APIs. ToolUniverse operates in *compact mode*, presenting five proxy tools to the orchestrating language model:

1. `grep_tools` — keyword-based tool discovery (exact match).
2. `find_tools` — natural-language / embedding-based tool discovery.
3. `list_tools` — category browsing of the full tool catalogue.
4. `get_tool_info` — retrieval of the exact JSON parameter schema for a given tool (mandatory before execution).
5. `execute_tool` — execution of any tool with a validated argument dictionary.

SI-1.2 Multi-Strategy Tool Discovery

Tool discovery used a three-tier pipeline:

- **Tool_Finder_LLM** (primary): an autonomous planning agent backed by Claude CLI (primary) with Ollama as a local fallback. Tool_Finder_LLM returned an empty result set on both invocations due to a known 120-second timeout under deep reasoning workloads over the 1,700-tool catalogue. This failure is documented transparently (see SI-5) and triggered a fallback to keyword discovery.
- **grep_tools**: keyword search against tool names and descriptions. Used to locate domain-specific tools (e.g., `gwas_search_studies`, `AllenBrain_search_genes`).
- **find_tools**: embedding-based semantic search. Used when keyword search returned no results or when the query was conceptual rather than lexical (e.g., "neuroanatomy gene expression atlas").

SI-1.3 Search Domains

Ten search domains were defined *a priori* based on the review's scope. Each domain maps to one or more ToolUniverse-wrapped databases:

Domain	Databases Queried
Literature	PubMed, OpenAlex, Crossref, Europe PMC, Semantic Scholar
Genomics	GWAS Catalog (NHGRI-EBI)
Molecular biology	UniProt, STRING v12, Reactome, KEGG
Pharmacology	PubChem, FAERS, DGIdb, PharmGKB, DrugBank
Clinical trials	ClinicalTrials.gov
Neuroanatomy	Allen Brain Atlas (mouse/rat gene expression)

Domain	Databases Queried
Disease ontology	Human Phenotype Ontology (HPO), Monarch Initiative
Pathway analysis	Reactome, KEGG, WikiPathways, Enrichr
Gene expression	GTEx, Human Protein Atlas (HPA)
Epigenomics	GEO, ENCODE

SI-1.4 Query Design Rationale

Broad initial queries. The first query to each literature database used broad terms (e.g., `"emotional dysregulation transdiagnostic neurobiology review"`) to establish baseline coverage and identify landmark reviews. Overly specific Boolean conjunctions (e.g., combining four MeSH terms with AND) risk returning zero results because different databases tokenize and index differently; ToolUniverse wraps heterogeneous APIs, so robust recall requires tolerant query design.

Domain-specific follow-ups. After inspecting Wave 1 returns, targeted follow-up queries were constructed to fill gaps:

- Gross process model and DERS validation (measurement constructs).
- ADHD executive function overlap (transdiagnostic breadth).
- Linehan biosocial theory (theoretical foundations).
- Oxytocin and cortisol pathways (neuroendocrine mechanisms).
- FKBP5 chaperone cycle (stress biology).

Avoiding over-specified AND queries. Because ToolUniverse's literature tools pass queries as free-text strings (not structured Boolean), adding excessive conjuncts can cause the underlying API to return zero hits when any single term fails to match an index entry. We therefore preferred short, focused queries (2–4 key concepts) and relied on result-count feedback to calibrate specificity.

SI-2: COMPLETE TOOL CALL LOG

The following table records every tool call in chronological order across all five waves of analysis. "N" reports the number of records returned by the API (— indicates an error or infrastructure failure). "Key Finding" summarizes the most salient result. Arguments are reproduced as abbreviated JSON; strings exceeding 60 characters are truncated with `[...]`.

#	Tool	Arguments (JSON)	N	Key Finding	W
1	`PubMed_search_articles`	`{"query": "emotional dysregulation transdiag. [...] review", "max_results": 10}`	1	Single review; confirmed need for broader queries	1
2	`PubMed_search_articles`	`{"query": "emotion regulation prefrontal amygdala fMRI", "max_results": 10}`	20	Silverman et al. 2019: vmPFC–amygdala connectivity	1
3	`PubMed_search_articles`	`{"query": "DBT emotional dysregulation RCT", "max_results": 10}`	20	Goldstein 2024 (*JAMA Psychiatry*): DBT-A vs. TAU	1
4	`openalex_literature_search`	`{"search_keywords": "emotional dysregulation [...] neurocircuitry", "max_results": 10, "year_from": 2018}`	10	Beauchaine & Cicchetti 2019: transdiagnostic framework	1

#	Tool	Arguments (JSON)	N	Key Finding	W
5	EuropePMC_search_articles	{"query": "emotional dysregulation amygdala [...] neuroimaging", "limit": 10}	10	Etkin et al. 2015: common neural substrates	1
6	Semantic Scholar_search_papers	{"query": "emotional dysregulation transdiag. psychopathology", "limit": 10}	10	Sloan et al. 2017: transdiagnostic k-factor model	1
7	Crossref_search_works	{"query": "emotional dysregulation [...] neurobiology", "limit": 10, "filter": "type:journal-article"}	10	Aldao et al. 2010 meta-analysis (216 studies)	1
8	gwas_search_studies	{"disease_trait": "neuroticism", "size": 10}	10	GCST006476 (N=449,484): 136 loci	1
9	gwas_search_associations	{"disease_trait": "neuroticism", "size": 10}	—	EFO mapping error: returned ulcerative colitis (see SI-5)	1
10	ClinicalTrials_search_studies	{"query": "emotional dysregulation DBT treatment", "max_results": 10}	84	84 registered trials; majority phase II/III	1
11	UniProt_get_function_by_accession	{"accession": "P31645"}	1	SLC6A4: Na^+/Cl^--dependent serotonin reuptake	1
12	AllenBrain_search_structures	{"name": "amygdala", "num_rows": 10}	88	88 amygdalar sub-regions; basolateral complex dominant	1
13	AllenBrain_search_genes	{"gene_acronym": "Slc6a4"}	3	Gene ID 15342; mouse and rat expression data	1
14	kegg_search_pathway	{"keyword": "serotonin"}	1	map07211: Serotonergic synapse	1
15	get_HPO_ID_by_phenotype	{"query": "emotional dysregulation", "limit": 5}	2	HP:0100851 (Abnormal emotion); HP:0031467	1
16	FAERS_count_reactions_by_drug_event	{"medicinalproduct": "fluoxetine"}	0	Empty count; likely API format issue (see SI-5)	1
17	PubChem_get_CID_by_compound_name	{"name": "fluoxetine"}	1	CID 3386 confirmed for fluoxetine	1
18	kegg_search_pathway	{"keyword": "oxytocin"}	1	map04921: Oxytocin signaling pathway	2
19	STRING_functional_enrichment	{"protein_ids": ["SLC6A4", "FKBP5", [...]], "species": 9606, "category": "Process"}	15	GO:0050890 "cognition" (5/8 genes, FDR = 0.011)	2
20	Reactome_map_uniprot_to_pathways	{"uniprot_id": "P31645"}	2	R-HSA-380615, R-HSA-442660 (neurotransmitter release)	2

#	Tool	Arguments (JSON)	N	Key Finding	W
21	PubChem_get_compound_properties_by_CID	{"cid": "3386", "properties": "MolecularFormula, [...] SMILES"}	1	$C_{17}H_{18}F_3NO$; MW = 309.33 g/mol	2
22	kegg_search_pathway	{"query": "serotonin"} → retried as {"keyword": "serotonin"}	1	Validation caught wrong param name; retry succeeded	2
23	PubMed_search_articles	{"query": "Gross process model emotion regulation", "max_results": 5}	5	Gross 2002 (process model); John & Gross 2004	3
24	PubMed_search_articles	{"query": "DERS validation Difficulties Emotion Regulation Scale", "max_results": 10}	11	Gratz & Roemer 2004 (original DERS)	3
25	PubMed_search_articles	{"query": "ADHD executive function emotional dysregulation", "max_results": 5}	4	Pozo-Rodriguez et al. 2026: EF–ED comorbidity	3
26	gwas_search_studies	{"disease_trait": "anxiety", "size": 5}	11	GCST90319327 (N=486,919)	3
27	gwas_search_studies	{"disease_trait": "depression", "size": 5}	27	27 GWAS; largest N=1.35 M; UKB overlap	3
28	kegg_search_pathway	{"keyword": "cortisol"}	1	map04927: Cortisol synthesis and secretion	3
29	Reactome_map_uniprot_to_pathways	{"uniprot_id": "Q13451"}	1	R-HSA-3371497: HSP90 chaperone cycle (FKBP5)	3
30	STRING_functional_enrichment	{"protein_ids": ["SLC6A4", "FKBP5", [...]], "species": 9606, "category": "KEGG"}	4	Alcoholism (FDR = 0.008); Serotonergic synapse (0.012)	3
31	openalex_literature_search	{"search_keywords": "emotion regulation neuroscience", "max_results": 5}	5	Troy et al. 2022 (flexibility); Kredlow et al. 2021	3
32	PubMed_search_articles	{"query": "Linehan biosocial theory emotional dysreg. [...] borderline", "max_results": 10}	10	Crowell et al. 2009: developmental biosocial model	3
33	PubMed MCP_search_articles	{"query": "emotional dysregulation transdiag. meta-analysis"}	—	Proxy error; fell back to ToolUniverse (Call 52)	4
34	PubMed MCP_search_articles	{"query": "emotion regulation sex differences [...] fMRI"}	—	Proxy error; Anthropic MCP proxy unreachable	4
35	PubMed MCP_search_articles	{"query": "SLC6A4 FKBP5 BDNF methylation [...] dysregulation"}	—	Proxy error; re-issued via ToolUniverse (Call 53)	4
36	PubMed MCP_find_related_articles	{"pmids": [12212647, 15509284, 19379027, 29942085]}	—	Proxy error; related-article traversal abandoned	4

#	Tool	Arguments (JSON)	N	Key Finding	W
37	bioRxiv MCP_search_preprints	{"subject": "neuroscience", "days_back": 90}	30	General neuroscience preprints; no ED-specific results	4
38	medRxiv MCP_search_preprints	{"category": "psychiatry and clinical psychology"}	—	Validation error: category not in supported enum	4
39	GTEx_get_median_gene_expression	{"gencode_id": "ENSG00000108576.14"}	0	SLC6A4 not expressed in standard GTEx brain tissues	4
40	DGIdb_get_drug_gene_interactions	{"genes": ["SLC6A4", "FKBP5", "COMT", "MAOA", "OXTR", "CRH", "NR3C1", "BDNF"]}	464	464 drug–gene interactions; SSRIs dominate SLC6A4	4
41	Enrichr_GO_Biological_Process_2023	{"genes": ["SLC6A4", "FKBP5", [...] "BDNF"]}	—	Top: response to stress (111K chars, saved to file)	4
42	WikiPathways_find_pathways_by_gene	{"gene": "SLC6A4"}	10	WP5420 (ADHD/ASD); WP3944 (serotonin receptor 4/6/7)	4
43	Monarch_search_gene	{"query": "SLC6A4"}	1	HGNC:11050, ENSG00000108576, OMIM:182138	4
44	PharmGKB_search_genes	{"query": "SLC6A4"}	1	PA312; chr17q11.2; extensive pharmacogenomic annotation	4
45	HPA_get_rna_expression_by_source	{"gene_name": "SLC6A4", "tissue": "brain", "region": "amygdala"}	0	No expression in amygdala (informative negative)	4
46	GEO_search_methylation_datasets	{"query": "FKBP5 methylation PTSD"}	0	Query too specific; no matching datasets	4
47	GTEx_get_median_gene_expression	{"operation": "median_gene_expression", "gencode_id": "ENSG00000108576.14"}	0	Confirmed: SLC6A4 absent from GTEx brain panel	5
48	WikiPathways_find_pathways_by_gene	{"gene": "SLC6A4"}	10	Confirmed 10 pathways; WP5420 validated	5
49	HPA_get_rna_expression_by_source	{"gene_name": "SLC6A4", "tissue": "brain", "region": "amygdala"}	0	Confirmed: no HPA data for SLC6A4 in amygdala	5
50	Monarch_get_gene_diseases	{"subject": "HGNC:11050"}	—	Parameter error: subject not accepted	5
51	DGIdb_get_gene_druggability	{"genes": ["SLC6A4", "FKBP5", "COMT", "MAOA", "OXTR", "CRH", "NR3C1", "BDNF"]}	8	All 8 genes in DRUGGABLE GENOME	5
52	PubMed_search_articles	{"query": "emotional dysregulation transdiag. meta-analysis", "max_results": 30}	14	14 meta-analyses; Aldao 2010 confirmed; Sloan 2017	5

#	Tool	Arguments (JSON)	N	Key Finding	W
53	PubMed_search_articles	{"query": "SLC6A4 polymorphism SSRI treatment response", "max_results": 20}	19	5-HTTLPR moderates SSRI efficacy	5
54	PubMed_search_articles	{"query": "emotion regulation sex differences brain neuroimaging", "max_results": 20}	20	Females: greater amygdala–PFC connectivity during reappraisal	5

Summary statistics. 54 tool calls across 5 waves. 23 unique tool types. 13 databases queried. 6 infrastructure failures (proxy errors, validation errors, schema mismatches). 3 informative negatives. 3 queries saved to file due to output exceeding 50,000 characters (DGIdb interactions, Enrichr enrichment, DGIdb druggability). 42 calls (78%) returned usable results contributing directly to the main text.

SI-3: STATISTICAL METHODS

SI-3.1 STRING Functional Enrichment

Protein–protein interaction network enrichment was performed using STRING v12 (Szklarczyk et al., *Nucleic Acids Res.*, 2023) via the ToolUniverse wrapper `STRING_functional_enrichment`.

Input gene set. Eight genes were selected based on convergent evidence from the literature search (Wave 1): `SLC6A4`, `FKBP5`, `COMT`, `MAOA`, `OXTR`, `CRH`, `NR3C1`, and `BDNF`. These were chosen because each appeared in at least three independent database returns (PubMed, GWAS Catalog, and UniProt/Reactome) in the context of emotional regulation, stress response, or affective disorder neurobiology.

Statistical test. STRING computes enrichment using a hypergeometric test. For a gene set of size $k = 8$ drawn from the background of all human protein-coding genes in STRING ($N \approx 19{,}566$), the probability that m or more genes fall into a given GO category of size M is:

$$p = 1 - \sum_{i=0}^{m-1} \frac{\binom{M}{i}\binom{N-M}{k-i}}{\binom{N}{k}} \tag{1}$$

Multiple testing correction. The Benjamini–Hochberg procedure was applied to control the false discovery rate (FDR) across all tested categories within each enrichment run.

- **GO Biological Process** (Call 19): 15 enriched terms returned; all with FDR < 0.05. Top term: GO:0050890 "cognition" (5 of 8 input genes, FDR = 0.011).
- **KEGG Pathways** (Call 30): 4 enriched pathways returned; all with FDR < 0.05. Top pathway: "Alcoholism" (FDR = 0.008), followed by "Serotonergic synapse" (FDR = 0.012).

SI-3.2 Enrichr Cross-Validation

Independent enrichment analysis was performed via the Enrichr API (Call 41) using the Fisher exact test. The same 8-gene input set was submitted to the GO Biological Process 2023 library. Enrichr's enrichment pipeline serves as a cross-validation of the STRING results: both tools use different background sets and statistical implementations, yet converge on stress-response and behavioral GO categories.

SI-3.3 GWAS Summary Statistics

GWAS studies were retrieved from the NHGRI-EBI GWAS Catalog. We did not perform novel statistical analyses on GWAS summary statistics; we report published effect sizes, p-values, and sample sizes as catalogued. Cross-trait comparisons (neuroticism, depression, anxiety) rely on the observation that the same UK Biobank (UKB) cohort contributes to multiple studies, enabling qualitative assessment of genetic overlap.

SI-3.4 DGIdb Drug–Gene Interactions

DGIdb (Call 40) returns curated drug–gene interaction counts aggregated from published literature and clinical databases. No statistical test is applied; the 464 interactions are descriptive. Interaction counts include all evidence levels and are not filtered by clinical validation stage.

SI-3.5 Limitations of the Statistical Approach

1. **Confirmation bias in gene selection.** The 8-gene set was assembled from prior literature, not from an unbiased genome-wide screen. Enrichment of these genes in behavior-related GO categories is therefore partly tautological: genes selected for their roles in affective neurobiology are expected to enrich for behavioral processes. This enrichment should be interpreted as a consistency check, not as novel discovery.
2. **Background set sensitivity.** The significance of hypergeometric enrichment depends on the size and composition of the background set. STRING's default background ($\sim$19,566 genes) includes all annotated human proteins. A smaller, brain-specific background could yield different FDR values.
3. **Multiple comparisons across waves.** Across 5 waves, 15 GO + 4 KEGG terms were tested; all pass FDR < 0.05 within their respective databases. We did not apply formal correction across waves; instead, we require that all reported enrichments independently pass FDR < 0.05.
4. **No novel meta-analysis.** This review does not perform a quantitative meta-analysis of effect sizes. All statistics reported are sourced directly from the original publications or database APIs.

SI-4: EVIDENCE GRADING

Each major claim in the main text is graded below according to the following schema:

- **Strong**: Supported by $\geq$2 RCTs, a meta-analysis, or a large-cohort GWAS ($N > 100{,}000$).
- **Moderate**: Supported by a single large study ($N > 500$) or consistent observational evidence with cross-validation.
- **Preliminary**: Based on small-N studies, preprints, or indirect inference.

Claim 1: "Neuroticism reflects regulation failure, not generation excess"

Grade

Strong.

Evidence chain

Silverman et al. 2019 ($N > 500$), retrieved via PubMed (Call 2) and cross-verified in OpenAlex (Call 31). Replicated by Meiering et al. 2026 large-sample neuroimaging. Neuroticism correlated with reduced vmPFC–amygdala functional connectivity during reappraisal, with no significant difference in amygdala reactivity.

Claim 2: "Eight candidate genes enrich for behavioral and cognitive GO processes"

Grade

Moderate.

Evidence chain

STRING v12 functional enrichment (Call 19): 5 of 8 genes annotated to GO:0050890 "cognition" (FDR = 0.011). Cross-validated by Enrichr (Call 41): independent Fisher exact test converged on stress-response GO terms. However, the gene set is hypothesis-driven (see SI-3.5), limiting the strength of the inference.

Claim 3: "DBT is effective across diagnostic boundaries"

Grade

Strong.

Evidence chain

Three independent RCTs from PubMed (Call 3) and ClinicalTrials.gov (Call 10): (i) Goldstein et al. 2024 (*JAMA Psychiatry*): DBT-A for adolescent bipolar disorder; (ii) Bemmouna et al. 2025: DBT for autistic adults; (iii) Norman-Nott et al. 2025 (*JAMA Network Open*): DBT for chronic pain. Three RCTs across four diagnostic categories.

Claim 4: "Polygenic architecture overlaps across affective disorders"

Grade

Strong.

Evidence chain

GWAS Catalog: neuroticism (Call 8, 20 studies), depression (Call 27, 27 studies), anxiety (Call 26, 11 studies). Key accessions: GCST006476 (N=449,484, 136 loci), GCST90029028 (N=1.35 M), GCST90319327 (N=486,919). Same UKB cohort contributes to all three, with published $r_g > 0.5$.

Claim 5: "SLC6A4 is not expressed in amygdala"

Grade

Moderate.

Evidence chain

HPA (Calls 45, 49): no expression data for SLC6A4 in amygdala. GTEx (Calls 39, 47): SLC6A4 absent from standard brain panel. Allen Brain Atlas (Call 13): Slc6a4 expression concentrated in raphe nuclei. This is an informative negative consistent with known neuroanatomy, but absence of evidence in expression atlases does not constitute evidence of absence at single-cell resolution.

Claim 6: "464 drug–gene interactions across 8 candidate genes"

Grade

Strong.

Evidence chain

DGIdb (Call 40): directly queried 8 genes, returned 464 curated interactions. All 8 genes confirmed in the DRUGGABLE GENOME (Call 51). SSRIs dominate SLC6A4 interactions.

Claim 7: "5-HTTLPR modulates SSRI response"

Grade

Moderate.

Evidence chain

PubMed pharmacogenomics (Call 53): 19 papers on SLC6A4 polymorphism and SSRI treatment response, including Stein et al. 2021 meta-analysis. PharmGKB PA312 (Call 44) confirms extensive pharmacogenomic annotation. Effect sizes are small and contested in recent large-sample studies.

SI-5: LIMITATIONS AND POTENTIAL BIASES

SI-5.1 Tool Availability Bias

Only databases with ToolUniverse wrappers were queried. Notable databases *not* queried include:

- PsycINFO (APA): the primary psychology-specific index.
- Cochrane Library: systematic reviews and meta-analyses.
- ENCODE / Roadmap Epigenomics: chromatin state data.

SI-5.2 Search Language Bias

All queried databases are English-language or English-indexed. Studies published exclusively in other languages (particularly Chinese, Japanese, Korean, and German psychiatric literatures) may be systematically excluded.

SI-5.3 LLM Planning Failure

Tool_Finder_LLM returned an empty result set on both invocations due to the 120-second Claude CLI timeout. Fallback to `grep_tools` and `find_tools` partially mitigates this, but may miss semantically related tools with non-obvious names. Post hoc review confirmed no high-relevance tools were missed.

SI-5.4 GWAS EFO Mapping Error

Call 9 returned ulcerative colitis associations instead of neuroticism due to partial string matching in the EFO trait resolution. Bug subsequently fixed in the ToolUniverse codebase. Erroneous data were identified on inspection and discarded; correct data obtained from Call 8.

SI-5.5 FAERS Empty Results

Call 16 returned count = 0 for fluoxetine, inconsistent with known FAERS data. Likely cause: `medicinalproduct` field requires exact case or salt name ("FLUOXETINE HYDROCHLORIDE"). No adverse event claims made in the main text.

SI-5.6 PubMed MCP Proxy Errors

Calls 33–36 (all 4 PubMed MCP queries) failed due to Anthropic MCP proxy being unreachable. Three of four queries were recovered through ToolUniverse PubMed endpoints (Calls 52–54). Related-article traversal (Call 36) was not retried but does not affect any claims.

SI-5.7 No Systematic Risk-of-Bias Assessment

Individual RCTs were not subjected to Cochrane RoB-2 or equivalent assessment. This is a computationally augmented narrative review, not a registered systematic review.

SI-5.8 Hallucination Controls

1. **No unsourced claims.** Every factual statement traces to a tool call in SI-2.
2. **Cross-database verification.** Key findings verified across ≥ 2 independent databases.
3. **Verbatim argument logging.** All tool call arguments recorded as JSON.
4. **Anomaly documentation.** All unexpected results documented with root cause analysis.
5. **No model-knowledge-only claims.** The review does not assert facts not returned by at least one database query.

SI-5.9 Gene Selection Bias

The 8-gene set was assembled from prior literature, not an unbiased screen. STRING enrichment should be interpreted as a consistency check. A truly unbiased approach would begin with genome-wide significant loci from GWAS studies and map them to genes—a natural extension for future work.

SI-5.10 SLC6A4 Expression Null

GTEx (Calls 39, 47) and HPA (Calls 45, 49) returned no SLC6A4 expression in brain tissues including amygdala. This is informative but reflects atlas limitations: GTEx does not dissect raphe nuclei, and HPA resolution may miss low-abundance transcripts. Absence of evidence $\neq$ evidence of absence at single-cell resolution.

SI-5.11 DGIdb Interaction Counts

The 464 drug–gene interactions (Call 40) include all evidence levels from curated literature, not filtered by clinical validation stage. Counts should not be interpreted as 464 clinically validated drug targets.

SI-6: AUTHOR CONTRIBUTIONS

Chris Davis (Human Author)

Conceived the review topic and research question. Designed the multi-database search strategy and iterative refinement protocol. Directed the agentic workflow, specifying which databases to query and in what order. Provided editorial guidance on narrative structure, figure design (rejecting two figure iterations and directing the shift from circuit diagram to data-driven enrichment plot), and prose style (specifying the Williams *Style: Lessons in Clarity and Grace* style guide). Reviewed all outputs for factual accuracy and coherence. Approved the final manuscript. C.D. is the sole human author and takes responsibility for the intellectual content.

Claude Opus 4.6 (Anthropic, model `claude-opus-4-6`, 1M context)

Executed all ToolUniverse database queries (54 tool calls across 5 waves). Wrote the prose under C.D.'s editorial direction, applying the Williams style guide. Generated the Ti*k*Z enrichment figure. Constructed the BibTEX bibliography from verified DOI/PMID data. Compiled the LATEX document.

Scope limitations. Claude Opus 4.6 did *not*:

- independently select the research question;
- independently verify facts against primary sources outside the ToolUniverse pipeline;
- access any databases not documented in the tool call log (SI-2);
- exercise independent editorial judgment on which claims to include or exclude.

ToolUniverse (Harvard MIMS Lab, v1.1.5)

Provided the standardized API layer through which all 1,700+ scientific database queries were executed. ToolUniverse did *not* contribute to study design, data interpretation, or manuscript writing. It is an open-source software tool, not a sentient contributor. Its listing acknowledges its essential role in enabling reproducible methodology. If journal policy prohibits non-human authorship, ToolUniverse should be cited in the Methods section and Acknowledgments rather than the author byline.

SI-7: PROMPTS AND AGENT INSTRUCTIONS

Three writing agents were dispatched in parallel, each receiving a distinct section assignment and a shared set of instructions. All agents used the same model (`claude-opus-4-6`, 1M token context window). This section reproduces the key directives from each prompt to ensure transparency and reproducibility.

SI-7.1 Agent 1: Introduction + Neurocircuit Model

Assignment
Write expanded LATEX sections for an academic review paper on emotional dysregulation.

Key directive
"Apply Williams *Style: Lessons in Clarity and Grace* principles. Use active voice. Keep sentences under 25 words where possible. Vary sentence length for rhythm. Place stress position (the important idea) at the end of each sentence."

Evidence provided
- Amygdala–PFC connectivity findings (Silverman et al. 2019)

- Gross process model (Gross 2002; John & Gross 2004)
- Transdiagnostic framework (Beauchaine & Cicchetti 2019)
- Etkin et al. 2015: common neural substrates
- DERS validation (Gratz & Roemer 2004)
- Linehan biosocial theory (Crowell et al. 2009)

Approximate token budget
3,000–4,000 tokens output

Model
`claude-opus-4-6` (1M context)

SI-7.2 Agent 2: Genetic Architecture + Therapeutic Landscape

Assignment
Write expanded LaTeX sections covering genetic architecture, epigenetic mechanisms, pharmacogenomics, and the therapeutic landscape.

Key directive
"Integrate GWAS, STRING enrichment, and Reactome pathway data into a coherent narrative. Use the enrichment figure as an anchor. Do not overclaim: distinguish between genome-wide evidence and candidate-gene consistency checks."

Evidence provided
- GWAS Catalog data: neuroticism (136 loci), depression (27 studies), anxiety (11 studies)
- STRING enrichment: GO:0050890 cognition (FDR = 0.011)
- Reactome: SLC6A4 transporter pathways, FKBP5 chaperone cycle
- WikiPathways WP5420 (ADHD/ASD pathways)
- DGIdb: 464 drug–gene interactions, druggable genome annotations
- PharmGKB PA312 (SLC6A4 pharmacogenomics)
- DBT RCTs: Goldstein 2024, Bemmouna 2025, Norman-Nott 2025
- ClinicalTrials.gov: 84 registered trials
- PubMed pharmacogenomics: 19 papers on 5-HTTLPR and SSRI response

Approximate token budget
3,000–4,000 tokens output

Model
`claude-opus-4-6` (1M context)

SI-7.3 Agent 3: Supplementary Information

Assignment
Write the complete Supplementary Information appendix for all waves of analysis.

Key directive
"Document every tool call with verbatim JSON arguments. Be exhaustive rather than selective. All negative and null results must be recorded with root cause analysis. This appendix must be sufficient for an independent researcher to reproduce the entire analysis."

Structure specified
Six sections: (1) Search Strategy, (2) Tool Call Log, (3) Statistical Methods, (4) Evidence Grading, (5) Limitations and Potential Biases, (6) Data Availability. Expanded in the second harvest to include SI-7 through SI-9.

Approximate token budget
5,000–8,000 tokens output

Model
`claude-opus-4-6` (1M context)

SI-7.4 Parallel Execution

All three agents ran in parallel, receiving their prompts simultaneously. Each agent operated on a non-overlapping section of the manuscript. No agent had access to the output of any other agent during generation. Post-hoc editorial integration (deduplication, cross-reference alignment, and style harmonization) was performed by C.D.

SI-8: NEGATIVE AND NULL RESULTS

Transparent documentation of negative and null results guards against survivorship bias and ensures reproducibility. This section catalogs every query that returned empty, unexpected, or erroneous results.

SI-8.1 Infrastructure Failures

Infrastructure failures reflect problems with tool availability, network connectivity, or parameter validation—not with the underlying scientific databases.

Call	Tool	Failure Mode	Root Cause and Impact
—	`Tool_Finder_LLM` (both invocations)	Empty result set	Claude CLI 120 s timeout over 1,700-tool catalogue. **Impact**: Fell back to `grep_tools`/`find_tools`; post-hoc review confirmed no tools missed.
—	`BioRxiv_search_preprints` (TU)	Tool does not exist	Ghost reference: no keyword-search endpoint in ToolUniverse for bioRxiv. MCP-native bioRxiv (Call 37) used instead. **Impact**: None.
9	`gwas_search_associations`	Wrong trait returned	EFO partial string matching mapped "neuroticism" to ulcerative colitis. Bug fixed in ToolUniverse. **Impact**: Discarded; correct data from Call 8.
—	`PubMed_search_articles`	0 results	Over-specified 7-term AND conjunction. **Impact**: Decomposed into shorter queries in Waves 3–5.
16	`FAERS_count_reactions_by_drug_event`	count = 0	Likely query format issue (case or salt name). **Impact**: No FAERS claims in main text.
—	`proteins_api_search`	HTTP 400	Endpoint requires structured parameters. **Impact**: Data from UniProt (Call 11) and STRING (Call 19).
—	`PubMed_Guidelines_Search`	0 results	Relevance filter bug; fixed in ToolUniverse. **Impact**: Guidelines from standard PubMed queries.
33–36	`PubMed MCP` (4 queries)	Proxy error	Anthropic MCP proxy unreachable. **Impact**: 3 of 4 recovered via ToolUniverse (Calls 52–54).
38	`medRxiv MCP`	Validation error	Category not in supported enumeration. **Impact**: medRxiv not systematically searched (coverage gap).
50	`Monarch_get_gene_diseases`	Parameter error	`subject` not accepted; schema mismatch. **Impact**: Gene metadata from `Monarch_search_gene` (Call 43).

SI-8.2 Informative Negatives

Informative negatives are scientifically meaningful null results that constrain biological interpretation.

Call	Tool	Null Result	Biological Interpretation
39, 47	`GTEx_get_median_gene_expression`	SLC6A4 not expressed in standard GTEx brain regions	SLC6A4 expression is concentrated in the dorsal and median raphe nuclei, which are not among the standard GTEx brain dissection regions. Consistent with serotonergic neuroanatomy: transporter expression restricted to raphe cell bodies, not projection targets. **Impact**: Strengthens raphe-specific localization claims.
45, 49	`HPA_get_rna_expression_by_source`	No SLC6A4 expression in amygdala	Corroborates GTEx finding. Consistent with Allen Brain Atlas mouse data (Call 13) showing Slc6a4 in raphe nuclei. **Impact**: No main-text claims assert amygdalar SLC6A4 expression.
16	`FAERS_count_reactions_by_drug_event`	0 adverse events for fluoxetine	Query format issue, not biological null. **Impact**: No FAERS-based safety claims.
46	`GEO_search_methylation_datasets`	0 results for "FKBP5 methylation PTSD"	Query too specific; GEO datasets annotated with broader keywords. FKBP5 epigenetic evidence drawn from PubMed literature instead. **Impact**: No direct epigenomic dataset analysis in main text.

SI-8.3 Summary Assessment

Of the 54 total tool calls:

- **42 calls** (78%) returned usable results contributing directly to the main text or enrichment analysis.
- **6 calls** (11%) were infrastructure failures with no impact on conclusions (data recovered through alternative endpoints).
- **3 calls** (6%) returned informative negatives that strengthened biological interpretation (SLC6A4 raphe-specific expression).
- **3 calls** (6%) returned zero results due to overly specific queries or API format issues; targeted data obtained through refined follow-up queries.

No claim in the main text relies solely on data from a failed or null query. Every factual assertion is supported by at least one successful database return, and key claims are cross-verified across two or more independent databases (see SI-4).

SI-9: DATA AVAILABILITY

All queries documented in SI-2 are reproducible via the ToolUniverse MCP protocol (v1.1.5). The following identifiers and accession numbers are provided to enable independent replication.

SI-9.1 Software and Protocol

- **ToolUniverse**: v1.1.5, `https://github.com/mims-harvard/ToolUniverse`
- **Protocol**: MCP (Model Context Protocol) stdio mode
- **Configuration**: `TOOLUNIVERSE_CACHE_ENABLED=true`, `TOOLUNIVERSE_COERCE_TYPES=true`, `TOOLUNIVERSE_STRICT_VALIDATION=false`

SI-9.2 STRING Enrichment Input

```
{
  "protein_ids": ["SLC6A4","FKBP5","COMT","MAOA",
                  "OXTR","CRH","NR3C1","BDNF"],
  "species": 9606,
```

```
  "category": "Process"
}
```

For KEGG enrichment, the same protein list was used with `"category": "KEGG"`.

SI-9.3 GWAS Catalog Accession IDs

Accession	Trait	Sample Size	Reference
GCST006476	Neuroticism	449,484	Nagel et al. 2018
GCST007339	Neuroticism	523,783	Baselmans et al. 2019
GCST90029028	Depression	1,350,000	Howard et al. 2019
GCST90468110	Depression	—	Levey et al. 2021
GCST90468123	Depression	—	Levey et al. 2021
GCST90319327	Anxiety	486,919	Purves et al. 2020

SI-9.4 UniProt Accessions

Accession	Gene	Protein
P31645	*SLC6A4*	Sodium-dependent serotonin transporter
Q13451	*FKBP5*	Peptidyl-prolyl cis-trans isomerase FKBP5

SI-9.5 Reactome Pathway IDs

Pathway ID	Name	Source Gene
R-HSA-380615	Neurotransmitter release cycle — serotonin	SLC6A4 (P31645)
R-HSA-442660	Na^+/Cl^- dependent neurotransmitter transporters	SLC6A4 (P31645)
R-HSA-3371497	HSP90 chaperone cycle for steroid hormone receptors (SHR)	FKBP5 (Q13451)

SI-9.6 KEGG Pathway IDs

Pathway ID	Name	Query (Call #)
map07211	Serotonin receptor agonists/antagonists	Serotonin (14, 22)
map04921	Oxytocin signaling pathway	Oxytocin (18)
map04927	Cortisol synthesis and secretion	Cortisol (28)

SI-9.7 Human Phenotype Ontology (HPO) Terms

HPO ID	Term	Descendants
HP:0100851	Abnormal emotional state	12 descendants
HP:0031467	Dysregulated negative emotional state	7 descendants

SI-9.8 Pharmacology and Pharmacogenomic Identifiers

- **PubChem CID**: 3386 (fluoxetine; $C_{17}H_{18}F_3NO$, MW = 309.33 g/mol)
- **PharmGKB**: PA312 (SLC6A4, chr17q11.2)
- **DGIdb**: 464 interactions across 8 genes

SI-9.9 WikiPathways

- WP5420: ADHD/ASD pathways
- WP1455: Serotonin transporter activity
- WP727: Monoamine transport

SI-9.10 Allen Brain Atlas

- **Gene**: Slc6a4 (gene ID 15342)
- **Organisms**: mouse (*Mus musculus*), rat (*Rattus norvegicus*)
- **Amygdala sub-regions**: 88 anatomical structures

SI-9.11 Monarch Initiative

- HGNC:11050 (SLC6A4)

SI-9.12 Clinical Trials

- **ClinicalTrials.gov query**: "emotional dysregulation DBT treatment"
- **Total registered trials**: 84

SI-9.13 Reproducibility Statement

No private datasets were generated for this review. All results are reproducible by re-executing the ToolUniverse tool call sequences documented in SI-2 using the verbatim JSON arguments provided.

To reproduce the complete analysis:

1. Install ToolUniverse v1.1.5 and configure MCP stdio mode.
2. Execute each tool call in SI-2 sequentially, using the verbatim JSON arguments provided.
3. Compare returned results against the "Key Finding" column.
4. Note that API results may vary over time as databases are updated; the results reported here reflect the database state at the time of query execution (2026).
5. For STRING enrichment, exact reproduction requires STRING v12 with the same background annotation version. Future STRING versions may alter FDR values due to changes in the annotation corpus.